Hyperspectral Image Classification via Cross-Domain Few-Shot Learning With Multilevel Style and Semantic Enhancement

Zhi Gong, Lijuan Duan, *Member, IEEE*, Fengjin Xiao, and Lei Tong, *Member, IEEE*

***Abstract*—One of the key challenges in cross-domain few-shot hyperspectral image classification (HSIC) lies in effectively leveraging spectral-spatial features while alleviating semantic inconsistencies between domains. To address this issue, we propose a multilevel style and semantic enhancement for cross-domain few-shot learning hyperspectral image classification (MSS-CFSL) framework, which simultaneously enhances low-level style features and high-level semantic representations to improve domain adaptation (DA). Specifically, MSS-CFSL dynamically aligns spatial and spectral features using 2-D spatial and 3-D spectral style information from the source domain (SD), ensuring better feature consistency in the target domain (TD). Furthermore, a learnable class-level shared representation is introduced to bridge the semantic gap, enhancing the model's ability to transfer discriminative knowledge under limited semantic annotations. In addition, MSS-CFSL integrates a spectral–spatial knowledge fusion mechanism with adaptive feature enhancement, balancing the extraction of structural and spectral band information. Extensive experiments on four hyperspectral datasets show that MSS-CFSL improves the overall accuracy (OA) on the four datasets by 11.15%, 12.82%, 5.04%, and 4.42%, respectively, when compared with eight state-of-the-art cross-domain few-shot methods, underscoring its effectiveness in cross-domain few-shot scenarios.**

***Index Terms*—Cross domain, few-shot learning (FSL), hyperspectral image classification (HSIC), multilevel feature.**

I. Introduction

THE hyperspectral imaging (HSI) is an advanced remote sensing technology that captures detailed spectral reflectance information across a wide range of wavelengths from surface objects. This capability allows for the identification and analysis of materials and substances that are otherwise difficult to distinguish using conventional imaging techniques. As a result, HSI has become invaluable in various applications, including precision agriculture, environmental monitoring, mineral exploration, and urban planning [1], [2], [3], [4], [5]. With the continuous advancement of deep learning, supervised learning-based HSI classification (HSIC) has achieved significant progress. Recently, models based on convolutional neural networks (CNNs) [6], [7], graph neural networks (GNNs) [8], [9], and the Transformer architecture [10], [11], [12] have demonstrated impressive results in HSIC when trained on a sufficient number of labeled samples in a single scene. However, the high spectral resolution of HSI necessitates expert knowledge for labeling, and the labeling process is costly, which limits its widespread application in practical scenarios.

To address the issue of limited labeled data in HSIC, domain adaptation (DA) [13], [14] has emerged as an effective solution, drawing increasing attention from the research community in recent years. The core idea of DA is to transfer knowledge from a source domain (SD) (with a sufficient number of labeled samples) to improve the model performance in a target domain (TD) (where labeled samples are scarce or even absent). Specifically, DA techniques help to mitigate the performance degradation caused by domain shift, enabling models to achieve better generalization in TD. Significant progress has been made in DA methods for HSI [15], [16], [17], [18], particularly with the advancement of adversarial learning, feature alignment, and other related techniques. However, as shown in Fig. 1, these methods generally require the SD and TD to share the same set of classes. When unseen classes appear, the performance of these models declines significantly, limiting their application in more complex, multiclass scenarios. Few-shot learning (FSL) [19] has emerged as a promising framework to address the issue of unseen classes in HSIC. FSL aims to learn new classes with a minimal number of labeled samples (see Fig. 1), which is crucial for the identification of emerging objects, unknown materials, and complex scenes in hyperspectral imagery. Unlike traditional deep learning approaches, FSL emphasizes the model's ability to rapidly adapt and generalize, enabling it to achieve high classification accuracy even with a small number of labeled samples. In recent years, FSL techniques based on meta-learning, data augmentation, and metric learning have been widely applied to various vision tasks, demonstrating competitive performance, particularly in the field of HSIC [20], [21], [22], [23], [24], [25].

Received 6 February 2025; revised 12 July 2025 and 20 September 2025; accepted 12 October 2025. Date of publication 21 October 2025; date of current version 31 October 2025. This work was supported in part by the Natural Science Foundation of Beijing under Grant L251052, in part by the National Natural Science Foundation of China under Grant 62472012, in part by Sichuan-Chongqing Science and Technology Innovation Cooperation Plan under Grant CSTB2024TIAD-CYKJCXX0029, in part by the Fengyun Satellite Ahead Pilot Program under Grant FY-APP-ZX-2023.02, and in part by Beijing University of Technology's "Urban Carbon Neutrality" and "Urban Renewal" Science and Technology Innovation Fund Projects. *(Corresponding author: Lijuan Duan.)*

Zhi Gong and Lijuan Duan are with Chongqing Research Institute, Beijing University of Technology, Beijing 100124, China, also with the Beijing Key Laboratory of Trusted Computing, Beijing 100124, China, also with the National Engineering Laboratory for Key Technologies of Information Security Level Protection, Beijing 100124, China, and also with the School of Computer Science, Beijing University of Technology, Beijing 100124, China (e-mail: gongzhi97@emails.bjut.edu.cn; ljduan@bjut.edu.cn).

Fengjin Xiao is with Beijing Climate Center, Beijing 100081, China.

Lei Tong is with the School of Computer Science, Beijing University of Technology, Beijing 100124, China.

Digital Object Identifier 10.1109/TGRS.2025.3624125

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

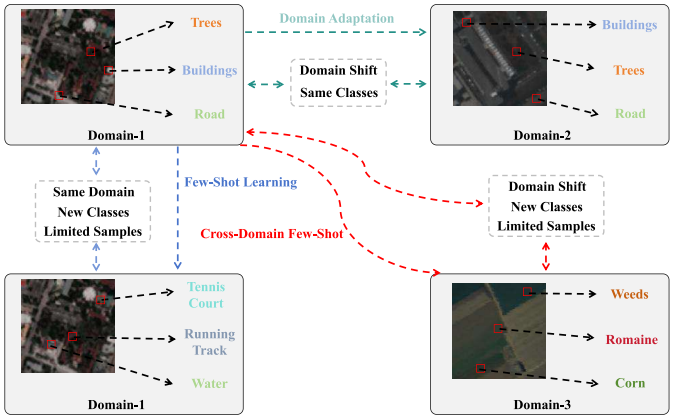


Fig. 1. Distinction among DA, FSL, and CDFSL. DA addresses the domain shift problem between the same classes across different domains; FSL focuses on learning new classes with limited labeled samples; CDFSL deals with the challenge of learning new classes in a TD with limited labeled samples, when data originate from a different SD.

Although FSL has demonstrated considerable success in addressing the problem of new class recognition with limited labeled samples, the inherent diversity of hyperspectral data—such as data distribution shifts caused by spectral band inconsistencies across heterogeneous sensors—poses significant challenges. As illustrated in Fig. 1, recent studies have increasingly focused on cross-domain FSL (CDFSL) to address both domain shift and the problem of limited labeled samples for new classes. Various approaches [26], [27], [28], [29], [30], [31], [32], [33], including feature alignment, meta-learning, and memory-based methods, have been proposed to address CDFSL. Feature alignment methods mitigate domain discrepancies through adversarial training or feature transformation, but they often underperform in the presence of complex domain shifts and fail to capture intricate semantic relationships in the hyperspectral data. Meta-learning methods emphasize rapid adaptation with limited data; yet, they demand meticulous task design and may struggle with large domain shifts or complex data. In contrast, memory-based methods, which preserve and leverage SD knowledge, demonstrate potential in enhancing TD performance. However, in HSIC, these methods frequently fall short in fully exploiting spectral and spatial features and fail to resolve semantic inconsistencies between SD and TD, which is critical for effective classification. These limitations highlight the need for more comprehensive and flexible solutions in CDFSL for HSIC.

To address this problem, we propose a novel approach to CDFSL-HSIC that addresses both the challenges of spectral–spatial feature exploitation and semantic inconsistencies across domains. Specifically, our method introduces a multilevel style and semantic enhancement framework that operates on both low-level and high-level features. Building on these insights, we propose a novel approach to CDFSL-HSIC that addresses both the challenges of spectral–spatial feature exploitation and semantic inconsistencies across domains. Specifically, our method introduces a multilevel style and semantic enhancement framework that operates on both low-level and high-level features. For low-level features, we leverage domain-specific knowledge in the form of 2-D spatial and 3-D spectral style features from SD to dynamically enhance the TD features, ensuring better alignment in both spatial and spectral dimensions. For high-level features, we introduce a learnable class-level shared representation that complements unique semantic features, effectively bridging the semantic gap between the SD and TD.

The main contributions of this work are summarized as follows.

1) We propose a novel approach for cross-domain few-shot HSIC: a multilevel feature enhancement framework that combines low-level dynamic style enhancement and high-level semantic feature optimization. Both components are designed to be network-agnostic, enabling seamless plug-and-play integration.
2) To fully exploit the unique characteristics of hyperspectral data, we propose a spectral–spatial knowledge integration method that dynamically enhances features by combining 2-D and 3-D network representations. Specifically, 2-D and 3-D networks capture spatial and spectral statistics for low-level features, enabling dynamic enhancement of TD features and improving spectral–spatial feature transferability.
3) To bridge the semantic gap between SD and TD in cross-domain tasks, we propose a class-level shared semantic representation mechanism. This mechanism introduces a learnable shared semantic feature to complement high-level TD features, thereby enhancing the semantic representation capacity of TD. It significantly improves the classification performance in few-shot tasks, particularly when semantic annotations are limited.
4) Our method not only significantly improves the transferability of both spectral and spatial features but also effectively addresses semantic discrepancies, thereby facilitating more accurate and robust classification in cross domain, FSL scenarios. Through extensive experiments, we demonstrate that our proposed approach consistently outperforms state-of-the-art methods, showcasing its exceptional effectiveness and adaptability in tackling the inherent challenges associated with the hyperspectral data analysis.

The remainder of this article is organized as follows. Section II briefly introduces HSIC. Section III describes the proposed multilevel style and semantic enhancement for cross-domain few-shot learning hyperspectral image classification (MSS-CFSL) in detail. Section IV gives the experimental results to verify the effectiveness of the proposed method. Finally, we conclude Section V.

II. Related Works

In this section, we review traditional learning methods for HSIC, deep learning approaches based on CNNs and other architectures, as well as CDFSL techniques for HSIC.

A. HSI Classification

HSIC is a critical task in remote sensing, enabling the identification and categorization of surface objects based on their unique spectral reflectance characteristics. Traditional methods for HSIC can be broadly categorized into several

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

groups, each with distinct mechanisms and limitations. Statistical model-based approaches, such as maximum likelihood classification (MLC) and Bayesian classification [34], rely on predefined assumptions about spectral feature distributions and classify pixels based on probabilistic measures. Distance-based techniques, including minimum distance classification and spectral angle mapper (SAM) [35], classify pixels by computing the distance or spectral angle relative to predefined class samples. Decision tree-based methods, such as random forest [36], leverage hierarchical structures or ensembles of decision trees to achieve classification [37]. Support vector machine (SVM) methods [38] determine an optimal hyperplane in a high-dimensional feature space to separate data classes. In addition, interspectral feature extraction techniques, such as principal component analysis (PCA) and independent component analysis (ICA) [39], improve classification performance by reducing dimensionality and eliminating redundant information. Neural network-based approaches, including multilayer perceptron (MLP) [40], harness the learning capabilities of neural networks to enhance classification outcomes. Despite their contributions, these traditional methods are often limited by rigid assumptions about data distributions, high computational costs, and reduced robustness in noisy or imbalanced datasets.

In recent years, deep learning has emerged as a powerful paradigm, addressing many of the limitations associated with traditional methods. CNNs, GNNs [9], [41], [42], and Transformer-based architectures [11], [43], [44] have demonstrated remarkable success in extracting spectral, spatial, and hybrid spectral–spatial features, significantly improving the classification accuracy and robustness. For example, Zhong et al. [45] proposed the spectral–spatial residual network (SSRN), an end-to-end 3-D CNN that integrates spectral and spatial residual blocks to effectively capture both types of information. Gong et al. [41] introduced a GNN-based approach that utilizes superpixels to extract features from spatial, spectral, and spectral–spatial dimensions, further enhancing the recognition accuracy. Inspired by the success of Transformers in natural language processing, recent studies have adapted Transformer-based models to HSIC. Notably, MorphFormer [44] combines spectral and spatial morphological networks with attention mechanisms to achieve superior classification performance. Similarly, WaveFormer [11] employs wavelet transforms for efficient, lossless downsampling, improving the integration of structural and shape information while maintaining a balance between classification performance and computational efficiency.

B. Cross-Domain Few-Shot Learning

Traditional classification approaches typically require a large amount of labeled data for training, which is both costly and time-consuming to acquire. FSL [19] addresses this challenge by enabling the recognition and classification of new categories with only a few labeled examples per class. For instance, Liu et al. [20] employed contrastive learning to construct a metric space that reduces sample uncertainty, while Wang et al. [30] introduced a spatial–spectral fusion network, later fine-tuned with transfer learning to extract discriminative spectral–spatial features. In addition, the integration of graph structures [21], [22], [23] into hyperspectral classification has facilitated feature extraction by aggregating neighborhood information. Notably, Wang et al. [21] extended the aggregation function to tasks involving rich small samples, combining the graph sample and aggregation network (GraphSAGE) with meta-learning into a unified framework. Recent advancements in FSL have further enhanced performance through attention mechanisms. For example, Ran et al. [24] proposed a spatial attention module and a spectral query module to reduce category uncertainty by leveraging information from distant samples, thereby improving the model's representational capacity. Similarly, Peng et al. [25] aligned the source and TDs to a common feature space using a domain aligner and employed a convolutional transformer (CT) network to extract both local and global features. Huang et al. [46] introduced a method for generating local spatial–spectral sequences based on spatial–spectral correlations and neighboring positional data, utilizing a Transformer-based network to explore spatial–spectral feature information from the input sequence.

Despite these advancements, FSL faces additional challenges when applied to new domains due to variations in data sources, such as differences in satellite sensors or geographical regions. In scenarios where SD and TD share identical categories, unsupervised DA techniques [26], [27], [28] are commonly employed to address domain shifts. However, when new categories emerge in the TD, CDFSL methods become essential. Deng et al. [29] pioneered the application of deep learning with unsupervised DA for cross-domain few-shot HSIC. Wang et al. [30] further improved classification performance by fine-tuning a spectral–spatial fusion network with VGGNet. Fang et al. [31] iteratively selected high-confidence target samples to reduce interdomain differences and enhance neural network performance. CMFSL [32] integrated global class representation learning, Mahalanobis distance-based label determination, a lightweight cross-scale convolutional network, and a spectral-prior-based refinement module to improve spectral–spatial information extraction and mitigate domain shifts. Gia-CFSL [33] combined FSL with domain alignment through graph information aggregation, utilizing IDE-block for intradomain relationships and CSA-block for cross-domain feature similarities. Recently, Huang et al. [47] proposed a novel CDFSL framework that employs batchwise pair considerations, kernel triplet loss for nonlinear relationship characterization, shared-feature extraction, and iterative adversarial training to effectively address domain shifts.

III. Proposed Method

The overall framework of the proposed MSS-CFSL is illustrated in Fig. 2. First, the HSI data from both the SD and TD undergo data preprocessing [extended morphological attribute profile (EMAP)] to construct input cubes, which are then divided into corresponding sets. The input features from both SD and TD are projected into a common dimensional space using distinct mapping methods. Subsequently, the spectral–spatial feature extractor derives semantic features for classification from the input data. During this process, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

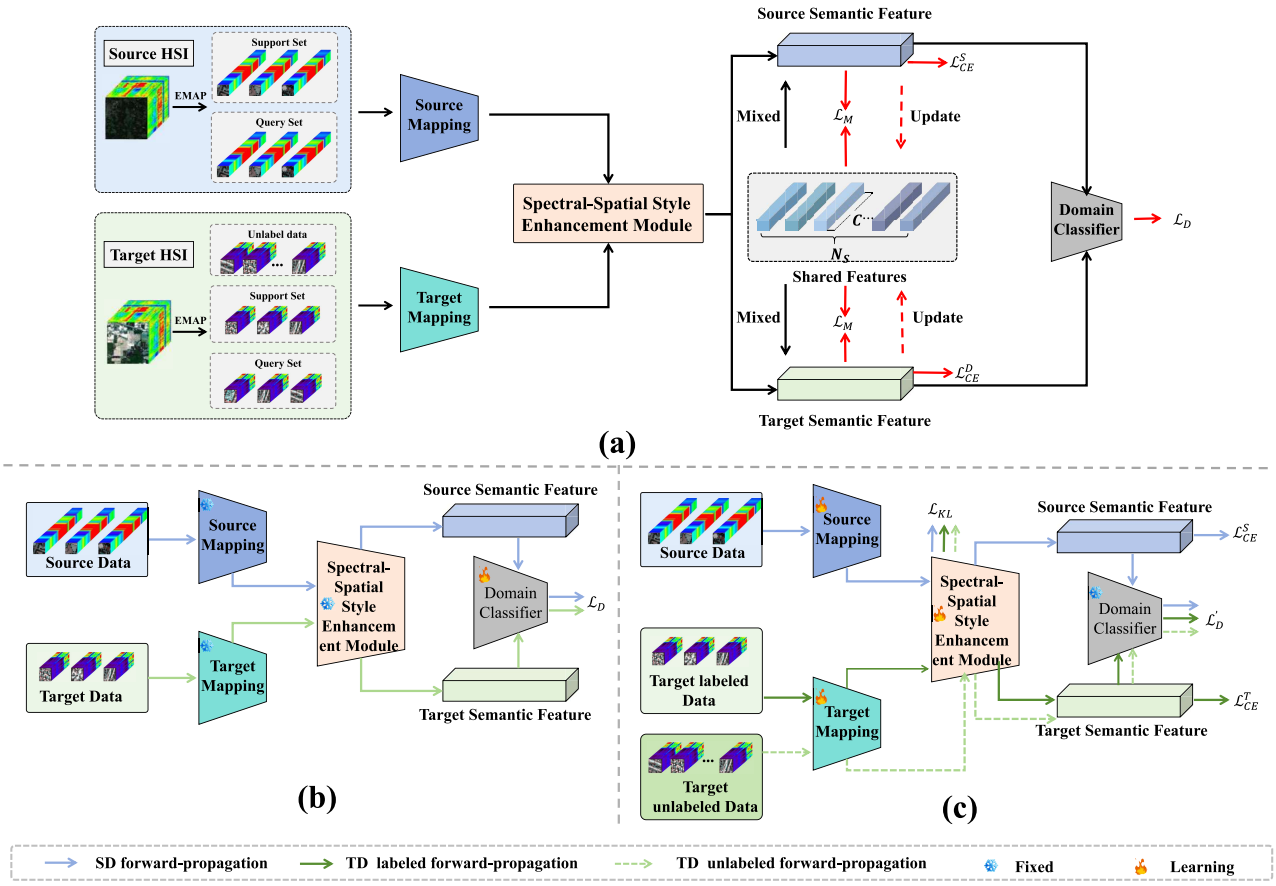


Fig. 2. (a) Overview framework of MSS-CFSL. (b) Training process of domain classifiers in MSS-CFSL. (c) Training process of feature extractor in MSS-CFSL. Overview of the MSS-CFSL method: we begin by preprocessing the data from SD and TD to obtain the input data. The SD and TD cubes are then projected into a common dimensional space through distinct mapping layers. Subsequently, the style information of SD is iteratively updated, and both SD and TD features are augmented. High-level semantic features are then extracted using a spectral–spatial style enhancement module, with trainable shared features employed to further augment the semantic information of SD and TD. In addition, we employ loss functions such as contrast loss, KL divergence, and mean square error in the domain classifier and feature extractor to limit the learning process of MSS-CFSL.

TD data are enhanced by leveraging the stored SD style knowledge, thereby significantly improving the transferability of the features. Finally, the extracted high-level semantic features are further refined through a learnable interclass shared features module, ultimately generating discriminative semantic features for classification.

A. Problem Setting

In this study, SD and TD denote HSI data acquired by different hyperspectral sensors in their respective domains. SD contains a sufficient number of labeled samples, while TD has only a limited number of labeled samples along with unlabeled samples to be predicted. Specifically, we formulate the CDFSL problem as follows. SD is defined as $D_S = \{(\boldsymbol{x}_s^i, \boldsymbol{y}_s^i) \mid i = 1, 2, \ldots, N_s\}$, where $\boldsymbol{x}_s^i$ and $\boldsymbol{y}_s^i$ represent the ith sample and its corresponding label, respectively. Similarly, the labeled samples in the TD are denoted by $D_T = \{(\boldsymbol{x}_t^i, \boldsymbol{y}_t^i) \mid i = 1, 2, \ldots, N_t\}$, and the unlabeled samples in TD are represented as $D_{T'} = \{\boldsymbol{x}_{t'}^i \mid i = 1, 2, \ldots, N_{t'}\}$. Here, $\boldsymbol{y}_s^i \in C_s$ and $\boldsymbol{y}_t^i \in C_t$, where C_s and C_t denote the set of classes in the source and TDs, respectively. In addition, N_s and N_t indicate the number of labeled samples in SD and TDs, respectively.

B. Framework

As depicted in Fig. 2, the proposed MSS-CFLS framework operates as follows: data from both SD and TD are first preprocessed. Next, distinct (nonshared) mapping layers and a shared-feature extraction network extract embedding features from the respective domains and align these features into a domain-invariant space.

The MSS-CFLS framework employs two adversarial processes that alternately train a domain classifier and a feature extraction network. As shown in Fig. 2(b), during the domain classifier training phase, parameters in both the mapping layer and feature extraction network remain frozen, while the domain classifier is updated. This stage exclusively optimizes the classifier to enhance discrimination between SD and TD. Inputs to the classifier consist of embedding features extracted from SD and TD samples. The classifier outputs probability values between 0 and 1, indicating sample origin (values closer to 1 represent a higher probability of SD origin). The loss $\mathcal{L}_D$ forces classifier outputs for SD samples toward 1 and those for TD samples toward 0.

Fig. 2(c) shows the process of training the feature extraction network. The domain classifier is frozen (parameters not updated), while the mapping layer and the feature extraction network are learnable (parameters updated). The embedding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

features of labeled samples from SD and TD are used to calculate losses to learn discriminant features. The Softmax outputs of labeled samples from SD and TD are used to calculate the losses L, respectively. The embedding features of unlabeled samples from the TD are also fed into the domain classifier to assist domain alignment. It should be noted that we only train the feature extraction network so that the outputs of the domain classifier for all samples (regardless of SD or TD origin) are expected to be as close to 1 as possible. Hence, the feature extraction network will fool the domain classifier so that it cannot distinguish whether the samples are from the SD or TD, i.e., the samples from SD and TD are mapped to a domain-invariant embedding feature space.

C. Spectral–Spatial Feature Extraction

First, we performed preprocessing on the HSI data from both the SD and TD. Adopting the preprocessing framework proposed by Huang et al. [47], we employed the EMAP technique, which has been empirically demonstrated to substantially enhance model accuracy. EMAP leverages multiple morphological attributes to more effectively model spatial information while significantly reducing the spectral dimensionality of HSI data. Following this, we constructed spatial–spectral patches centered on each pixel using the extracted EMAP features, organizing them into 3-D cubes to serve as the final preprocessed HSI data for both SD and TD.

In the feature extraction stage, to address the challenge of cross-domain few-shot transfer learning, we propose a dual knowledge base framework based on style feature alignment (as shown in Fig. 3), named 2-D style alignment and dynamic adjustment (SADA_V2) and 3-D style alignment and dynamic adjustment (SADA_V3). This framework stores the low-level feature distribution information of SD (represented by mean and variance) to enhance the feature representation capability of TD and reduce the domain discrepancy. Specifically, we decompose the low-level style features into spatial distribution differences (e.g., illumination, color, and texture) and spectral distribution differences. During feature extraction, the SADA_V2 module employs a 2D-CNN to extract spatial features, while the SADA_V3 module utilizes a 3D-CNN to extract joint spectral–spatial features. This design enables the model to better adapt to the diverse low-level style features stored in SD, thereby achieving more effective cross-domain feature alignment and transfer.

1) Spatial Feature Extraction: To extract spatial features from HSI, we employ a 2D-CNN, which excels at capturing spatial patterns and hierarchical spatial features through convolutional operations. Specifically, we construct multilevel pooling features and utilize max pooling to extract salient features. Max pooling not only mitigates the influence of noise but also reduces interdomain discrepancies by aligning the prominent features between SD and TD

$$F_s = \text{Concat}\left\{\text{MaxPool}_i\left(F_{s_i}\right)\right\} \tag{1}$$

where Concat$(\cdot)$ represents the feature concatenation operation and MaxPool$_i(\cdot)$ represents the maximum pooling operations at different levels.

2) Spectral Feature Extraction: To extract spectral features from HSI, we utilize a 3-D-CNN, which effectively captures spectral continuity and enables multidimensional feature fusion. By employing mean pooling operations across both spectral and spatial dimensions, we enhance the integration of spatial–spectral information, significantly improving the discriminative power of the extracted features

$$F_{\text{sp}} = \text{MeanPool}_{3-\text{D}}\left(\text{ReLU}\left(\text{Conv3D}_i\left(\left(F_{\text{sp}_i}, W_i\right) + b_i\right)\right)\right) \tag{2}$$

where MeanPool$_{3-\text{D}}(\cdot)$ represents a 3-D mean pooling operation, ReLU$(\cdot)$ is the activation function, and Conv3D$_i$ represents a 3-D convolutional operation.

To fully leverage the complementary strengths of spatial and spectral features, we employ a weighted averaging strategy for feature fusion

$$F_{\text{mix}} = \gamma F_s + (1-\gamma) F_{\text{sp}} \tag{3}$$

where F_s is the spatial feature extracted by 2D-CNN, F_{sp} is the spectral feature extracted by 3-D-CNN, and F_{mix} is the mixed feature after fusion. γ is the learnable weight coefficient.

To ensure the fused features preserve the discriminative information of both spatial and spectral domains, we constrain their distribution to closely align with the original spatial and spectral feature distributions. To achieve this, we introduce the Kullback–Leibler (KL) divergence as a regularization term during the feature fusion process

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}\left(F_{\text{mix}} \| F_s\right) + D_{\text{KL}}\left(F_{\text{mix}} \| F_{\text{sp}}\right). \tag{4}$$

D. Style Alignment and Dynamic Adjustment (SADA)

The proposed SADA framework is designed to address the challenge of cross-domain transferability in HSIC. SADA achieves this by aligning the low-level feature distributions between SD and TD, thereby enhancing the feature representation capability of TD. Specifically, SADA decomposes the low-level style features into spatial distribution attributes and spectral distribution attributes, which are critical for capturing the unique characteristics of HSI data. To implement this, SADA employs a dual-branch architecture: the SADA_V2 module utilizes a 2D-CNN to extract and align spatial features, while the SADA_V3 module leverages a 3-D-CNN to extract and align joint spectral–spatial features.

1) SADA_V2: The SADA_V2 module aligns spatial features between the SD and TD by leveraging low-level style representations extracted via a 2D-CNN. In HSIC, spatial features are essential for encoding domain-specific attributes such as texture and illumination. SADA_V2 constructs a spatial knowledge base by iteratively storing and adapting spatial distribution information from SD, effectively reducing domain discrepancies and dynamically refining the TD's feature representation. Inspired by [48] and [49], which demonstrate that feature means and variances can capture domain-invariant characteristics, we initialize N_{spa} sets of means $M_{\text{spa}} = \{\mu_{\text{spa}}^0, \mu_{\text{spa}}^1, \ldots, \mu_{\text{spa}}^{N_{\text{spa}}-1}\}, \mu_k \in \mathbb{R}^{C_l}$ and variances $\Sigma_{\text{spa}} = \{\sigma_{\text{spa}}^0, \sigma_{\text{spa}}^1, \ldots, \sigma_{\text{spa}}^{N_{\text{spa}}-1}\}, \sigma_k \in \mathbb{R}^{C_l}$ to represent diverse distribution patterns, thereby constructing the spatial knowledge base. C_l represents the number of feature channels in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

lth layer, and k represents the kth mean and variance. Then, M_{spa} and Σ_{spa} form N_{spa} pairs of knowledge bases for storing different feature distributions.

In the SADA_V2 module, we compute the mean and variance of spatial features from the SD to capture their distribution characteristics. Specifically, for a given spatial feature f^s_{spa}, the mean μ^s_{spa} and variance σ^s_{spa} are calculated as follows:

$$\mu^s_{\text{spa}} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} f^{s^{c,h,w}}_{\text{spa}} \tag{5}$$

$$\sigma^s_{\text{spa}} = \sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(f^{s^{c,h,w}}_{\text{spa}} - \mu^s_{\text{spa}}\right)^2} \tag{6}$$

where H and W denote the size of the feature map.

To fully store the SD information, we adopt a similarity accumulation strategy to dynamically write the mean and variance of SD into the knowledge base. Specifically, we use cosine similarity to calculate the similarity between the current mean μ^k_{spa} of knowledge base and the mean μ^s_{spa} of SD, denoted by $\cos(\mu^k_{\text{spa}}, \mu^s_{\text{spa}})$, as well as the similarity between the current variance σ^k_{spa} and the variance σ^s_{spa} of SD, denoted by $\cos(\sigma^k_{\text{spa}}, \sigma^s_{\text{spa}})$. During training, we normalize the similarity scores between current SD knowledge and each distribution in the knowledge base, utilizing the normalized values as weights for knowledge integration. Following the integration of SD knowledge, the mean M_{spa} and variance Σ_{spa} in the knowledge base are updated as follows:

$$\mu^k_{\text{spa}} = \alpha\mu^k_{\text{spa}} + (1-\alpha)\frac{\cos\left(\mu^k_{\text{spa}}, \mu^s_{\text{spa}}\right)}{\sum_{k=1}^{N_{\text{spa}}}\cos\left(\mu^k_{\text{spa}}, \mu^s_{\text{spa}}\right)}\mu^s_{\text{spa}} \tag{7}$$

$$\sigma^k_{\text{spa}} = \alpha\sigma^k_{\text{spa}} + (1-\alpha)\frac{\cos\left(\sigma^k_{\text{spa}}, \sigma^s_{\text{spa}}\right)}{\sum_{k=1}^{N_{\text{spa}}}\cos\left(\sigma^k_{\text{spa}}, \sigma^s_{\text{spa}}\right)}\sigma^s_{\text{spa}} \tag{8}$$

where α represents the update weight of the knowledge base.

According to (7) and (8), the spatial knowledge of SD is iteratively integrated into the knowledge base. To ensure that each pair of means and variances in the knowledge base effectively represents distinct knowledge, we enforce an orthogonality constraint by minimizing their pairwise cosine similarity, driving it toward 0

$$\mathcal{L}^{\text{diff}}_{\text{spa}} = \frac{1}{{N_{\text{spa}}}^2}\sum_{i=1}^{N_{\text{spa}}}\sum_{j=1}^{N_{\text{spa}}}\frac{\mu^i_{\text{spa}}\cdot\mu^j_{\text{spa}}}{\|\mu^i_{\text{spa}}\|\|\mu^j_{\text{spa}}\|} + \frac{\sigma^i_{\text{spa}}\cdot\sigma^j_{\text{spa}}}{\|\sigma^i_{\text{spa}}\|\|\sigma^j_{\text{spa}}\|}. \tag{9}$$

Equation (9) ensures the independence of the SD knowledge learned by the knowledge base.

SADA_V2's spatial knowledge base stores a large amount of independent spatial style knowledge information of SD. To bridge the domain gap and enhance the model's generalization ability, we leverage the information in this knowledge base to dynamically enhance the feature representations of SD and TD. By integrating the stored spatial style information, we aim to align the feature distributions of SD and TD, thereby reducing the domain gap and improving the model's adaptability to diverse domain features. To further enhance the feature representation, we introduce an attention mechanism,

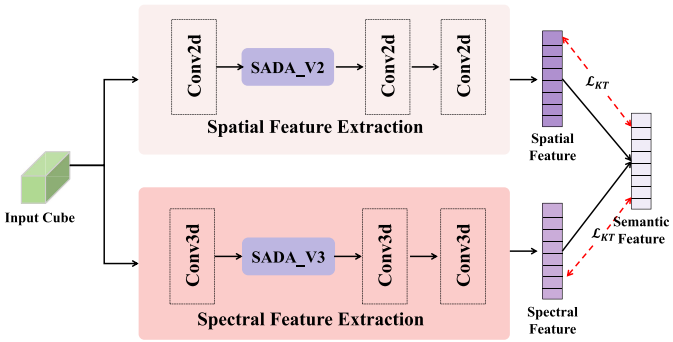


Fig. 3. Spectral–spatial style enhancement module encompasses both spectral feature extraction and spatial feature extraction.

where the information in the spatial knowledge base serves as keys and values, while the information from SD or TD acts as the query. This attention mechanism dynamically computes the relevance between the query and the entries in the knowledge base, assigning higher weights to the most relevant spatial style information. By aggregating the weighted values, an enhanced feature representation is generated

$$\mu^{\text{aug}}_{\text{spa}} = \text{Softmax}\left(\frac{\mu_{\text{spa}}{M_{\text{spa}}}^T}{\sqrt{d_k}}\right)M_{\text{spa}} \tag{10}$$

$$\sigma^{\text{aug}}_{\text{spa}} = \text{Softmax}\left(\frac{\sigma_{\text{spa}}{\Sigma_{\text{spa}}}^T}{\sqrt{d_k}}\right)\Sigma_{\text{spa}} \tag{11}$$

where the Softmax function serves as a normalization function and $(d_k)^{1/2}$ acts as a scaling factor to stabilize the gradient during training.

Next, SADA_V2 reconstructs the standardized spatial features f_{spa} using the enhanced mean $\mu^{\text{aug}}_{\text{spa}}$ and variance $\sigma^{\text{aug}}_{\text{spa}}$. This process maps the features back to a new distribution space, leveraging the enhanced statistical properties to improve feature representation and domain alignment

$$f^{\text{aug}}_{\text{spa}} = \sigma^{\text{aug}}_{\text{spa}}\frac{f_{\text{spa}} - \mu_{\text{spa}}}{\sigma_{\text{spa}}} + \mu^{\text{aug}}_{\text{spa}}. \tag{12}$$

In this process, the features of both TD and SD are dynamically enhanced through the spatial knowledge base, which stores a rich collection of SD spatial style information. As training progresses, this enhancement mechanism is continuously refined, significantly improving the generalization capability of the features and effectively reducing the domain discrepancy. The visualization of this process in SADA_V2, as shown in (see Fig. 4), demonstrates the dynamic integration of the knowledge of SD and the utilization of the knowledge base for adaptive feature enhancement.

2) SADA_V3: While SADA_V2 effectively enhances spatial features through its spatial knowledge base, the integration of spectral information is equally critical in HSIC. Hyperspectral data inherently contain rich spectral–spatial characteristics, and relying solely on spatial features may overlook the discriminative power embedded in the spectral domain. Therefore, we introduce SADA_V3, a complementary module that focuses on extracting and aligning spectral–spatial features using a 3-D-CNN. By leveraging both spatial and spectral information, SADA_V3 ensures a more comprehensive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

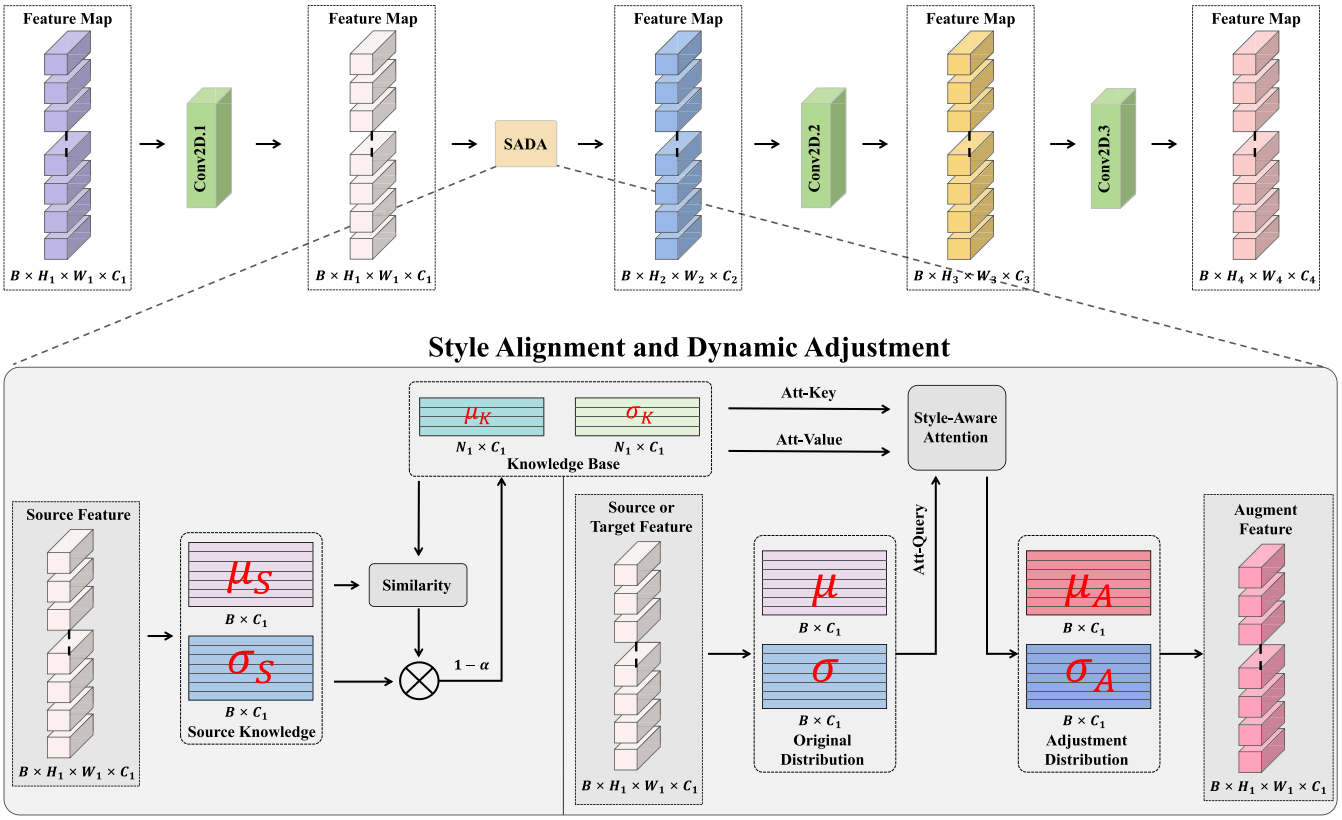


Fig. 4. Specific details of SADA.

representation of the data, further reducing domain discrepancies and enhancing the model's generalization capability in cross-domain few-shot scenarios. The specific workflow of SADA_V3 is also illustrated in Fig. 4, with the distinction that it operates on a 3-D-CNN to enhance the representation of spectral features $f_{\text{spe}} \in \mathbb{R}^{B\times C'\times D\times H\times W}$.

Similarly, in SADA_V3, we initialize N_{spe} sets of means $M_{\text{spe}} = \{\mu_{\text{spe}}^0, \mu_{\text{spe}}^1, \ldots, \mu_{\text{spe}}^{N_{\text{spe}}-1}\}$, where $\mu_k \in \mathbb{R}^{C'_l}$, and variances $\Sigma_{\text{spe}} = \{\sigma_{\text{spe}}^0, \sigma_{\text{spe}}^1, \ldots, \sigma_{\text{spe}}^{N_{\text{spe}}-1}\}$, where $\sigma_k \in \mathbb{R}^{C'_l}$, to represent diverse distribution patterns. This initialization forms the foundation of the spectral knowledge base, enabling the model to capture and adapt to the spectral characteristics of the SD, thereby enhancing the feature representation and domain alignment capabilities in HISC.

In SADA_V3, we employ the same methodology as SADA_V2 to integrate the spectral information from the SD into the spectral knowledge base

$$\mu_{\text{spe}}^k = \alpha\mu_{\text{spe}}^k + (1-\alpha)\frac{\cos\left(\mu_{\text{spe}}^k, \mu_{\text{spe}}^s\right)}{\sum_{k=1}^{N_{\text{spe}}}\cos\left(\mu_{\text{spe}}^k, \mu_{\text{spe}}^s\right)}\mu_{\text{spe}}^s \tag{13}$$

$$\sigma_{\text{spe}}^k = \alpha\sigma_{\text{spe}}^k + (1-\alpha)\frac{\cos\left(\sigma_{\text{spe}}^k, \sigma_{\text{spe}}^s\right)}{\sum_{k=1}^{N_{\text{spe}}}\cos\left(\sigma_{\text{spe}}^k, \sigma_{\text{spe}}^s\right)}\sigma_{\text{spe}}^s \tag{14}$$

where μ_{spe}^s and σ_{spe}^s are computed from the low-level spectral features f_{spe} of SD.

For the spectral features of both SD and TD, we utilize an attention mechanism to enhance the samples by leveraging the information from the spectral knowledge base, thereby reducing the spectral discrepancy between SD and TD

$$f_{\text{spe}}^{\text{aug}} = \left(\text{Softmax}\left(\frac{\sigma_{\text{spe}}\Sigma_{\text{spe}}{}^T}{\sqrt{d_k}}\right)\Sigma_{\text{spe}}\right)\frac{f_{\text{spe}}-\mu_{\text{spe}}}{\sigma_{\text{spe}}} + \text{Softmax}\left(\frac{\mu_{\text{spe}}M_{\text{spe}}{}^T}{\sqrt{d_k}}\right)M_{\text{spe}}. \tag{15}$$

Furthermore, SADA_V3 imposes an orthogonality constraint on its N_{spe} sets of spectral information to ensure mutual independence among the stored spectral patterns

$$\mathcal{L}_{\text{spe}}^{\text{diff}} = \frac{1}{N_{\text{spe}}{}^2}\sum_{i=1}^{N_{\text{spe}}}\sum_{j=1}^{N_{\text{spa}}}\frac{\mu_{\text{spe}}^i\cdot\mu_{\text{spe}}^j}{\|\mu_{\text{spe}}^i\|\|\mu_{\text{spe}}^j\|} + \frac{\sigma_{\text{spe}}^i\cdot\sigma_{\text{spe}}^j}{\|\sigma_{\text{spe}}^i\|\|\sigma_{\text{spe}}^j\|}. \tag{16}$$

By integrating both spectral and spatial style constraints, we enhance the diversity and discriminative power of low-level style features, thereby improving DA and classification performance

$$\mathcal{L}^{\text{diff}} = \mathcal{L}_{\text{spe}}^{\text{diff}} + \mathcal{L}_{\text{spa}}^{\text{diff}}. \tag{17}$$

E. Learnable Class-Level Shared Features (LCSFs)

While the low-level dual knowledge bases in SADA_V2 and SADA_V3 effectively capture spatial and spectral style information, they primarily focus on enhancing domain-invariant features at the lower levels. To further enrich the semantic representation and improve the discriminative power of high-level features, we introduce an LCSF module, as shown in Fig. 5. This module complements the unique semantic characteristics of each class by learning a shared representation that encapsulates common patterns across domains.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

Since there is no class overlap between SD and TD, shared features must be capable of spanning all classes while remaining domain-agnostic. To effectively learn these shared features, we define N_s learnable feature vectors f_{sh}. These vectors f_{sh} continuously interact with high-level semantic features $f_{\text{high}} \in \mathbb{R}^{HW\times C}$ to capture the relevant feature information, ensuring a robust and generalizable representation for cross-domain few-shot classification tasks.

To facilitate the interaction between high-level semantic features f_{high} and shared features f_{sh}, we compute the similarity between f_{high} and f_{sh} to evaluate the alignment of each feature with the shared representation. Based on this similarity, we generate a new visual feature $f_{\text{high}}^{\text{aug}}$ that integrates shared information between classes

$$f_{\text{high}}^{\text{aug}} = \left(\left(\frac{\left(f_{\text{high}}\right) {f_{\text{sh}}}^{T}}{\| \left(f_{\text{high}}\right) \| \| f_{\text{sh}} \|} \right)^{T} f_{\text{sh}} \right). \tag{18}$$

To ensure that the generated visual feature $f_{\text{high}}^{\text{aug}}$ retains the original high-level semantic information of f_{high} while incorporating class-shared information, we impose a mean squared error (mse) constraint between $f_{\text{high}}^{\text{aug}}$ and f_{high}. This constraint preserves the class-specific characteristics essential for accurate classification, while simultaneously integrating shared patterns across classes to enhance the model's generalization capability

$$\mathcal{L}_{M_1} = \frac{1}{HW} \sum_{i=1}^{HW} \left(f_{\text{high}}^{\text{aug}\,(i)} - {f_{\text{high}}}^{(i)} \right)^2. \tag{19}$$

In an ideal scenario, LCSF are capable of reconstructing certain characteristics of the semantic features. To achieve this, we leverage the shared features to reconstruct the feature f_{re}, while imposing a constraint to ensure its alignment with the original semantic feature f_{high}. This constraint guarantees the correctness of the shared features, ensuring that they accurately capture and represent shared patterns across classes

$$\mathcal{L}_{M_2} = \frac{1}{HW} \sum_{i=1}^{HW} \left(f_{\text{re}}^{(i)} - {f_{\text{high}}}^{(i)} \right)^2 \tag{20}$$

$$\mathcal{L}_M = \mathcal{L}_{M_1} + \mathcal{L}_{M_2}. \tag{21}$$

By incorporating the aforementioned constraints, we ensure that the final feature $f_{\text{high}}^{\text{aug}}$ integrates both high-level semantic features and shared features. This integration enables the model to learn richer and more accurate semantic representations, significantly enhancing its performance in cross-domain few-shot classification tasks.

F. Loss Function

In the cross-domain few-shot task training phase, the domain discriminator and feature extractor are trained alternately to optimize their respective objectives.

1) Training Domain Classifier: As shown in Fig. 2(b), during the training of the domain classifier, we first freeze the parameters of the mapping layer and feature extractor and only update the parameters of the domain classifier. The domain classifier needs to learn to distinguish the features of SD and TD. We set the domain label of SD to 1 and the domain label of TD to 0. The domain classifier is constrained by the binary cross-entropy loss $\mathcal{L}_D$ function so that the output of the SD feature is close to 1 and the output of the TD feature is close to 0

$$\mathcal{L}_D = \frac{1}{N'} \sum_{i=1}^{N'} \left[y_i^D \log D\left(f_i\right) + \left(1 - y_i^D\right) \log\left(1 - D\left(f_i\right)\right) \right] \tag{22}$$

where N' is the total number of samples, y_i^D is the domain label of the ith sample, and $D(f_i)$ is the domain classifier output of the ith sample.

2) Training Feature Extraction Network: As shown in Fig. 2(c), during the training of the feature extractor, we begin by freezing the parameters of the domain classifier while updating those of the mapping layer and the feature extractor. Throughout this process, features are extracted for classification using both SD data and labeled TD data, with constraints imposed by cross-entropy loss $\mathcal{L}_{\text{CE}}^S$ and $\mathcal{L}_{\text{CE}}^T$. In addition, unlabeled TD data are incorporated to aid in aligning domain features. By strategically misleading the domain classifier to hinder its ability to accurately differentiate between SD and TD features, we anticipate that the output of SD and TD features after passing through the domain classifier will tend closer toward 1.

The cross-entropy losses $\mathcal{L}_{\text{CE}}^S$ and $\mathcal{L}_{\text{CE}}^T$ are defined as follows:

$$\mathcal{L}_{\text{CE}}^S = -\frac{1}{N_S} \sum_{i=1}^{N_S} y_i^s \log \hat{y}_i^s \tag{23}$$

$$\mathcal{L}_{\text{CE}}^T = -\frac{1}{N_T} \sum_{i=1}^{N_T} y_i^t \log \hat{y}_i^t \tag{24}$$

where N_S is the number of data in SD and N_T is the number of labeled data in TD. y_i is the true label of the sample, and $\hat{y}_i$ is the predicted probability value of the sample.

The overall loss used to train the network is

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CE}}^S + \mathcal{L}_{\text{CE}}^T + \lambda_D \mathcal{L}_D + \lambda_K \mathcal{L}_{\text{KL}} + \lambda_d \mathcal{L}_{\text{diff}} + \lambda_M \mathcal{L}_M \tag{25}$$

where λ_D, λ_K, λ_d, and λ_M are to balance the weights between losses. λ_D, λ_K, λ_d, and λ_M are 0.01, 1, 0.01, and 1, respectively. $\mathcal{L}_{\text{diff}}$ and $\mathcal{L}_M$ are defined in the following.

The algorithm flow of the final training phase is shown in Algorithm 1.

IV. Experiments

A. Datasets

We utilized five HSI datasets to validate our approach, namely, Chikusei, Indian Pines (IP), University of Pavia (UP), Salinas, and Houston (HO). In the experiments, Chikusei served as the SD dataset, while the other four were used as TD datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 MSS-CFSL

Input: Source data set D_S, Target data set D_T and $D_{T'}$, Learning Rate η, Number of iterations EPOCHS, Loss Weight

Output: Domain Classifier and Feature Extraction Network

Initialize the mean M and variance Σ in the low-level style feature module ;
Initialize the shared-feature f^s;
Initialize network parameters;
for *each epoch* **do**
  Data Sampling;
  $D_S = (\boldsymbol{x}_s^i, \boldsymbol{y}_s^i)$, $D_T = (\boldsymbol{x}_t^i, \boldsymbol{y}_t^i)$, $D_{T'} = \{\boldsymbol{x}_{t'}^i\}$;
  Extracting features through two mapping layers;
  Update the Spatial Knowledge Base and Spectral Knowledge Base;
  Augmenting low-level features f_{spa}^{aug} and f_{spe}^{aug} using stylized information according to eq. (12) and eq. (15);
  Extracting spectral–spatial features F_{mix} according to eq. (3);
  Augmenting high-level semantic features using shared-feature f_{high}^{aug} according to eq. (18) ;
  Training Domain Classifier: compute domian loss $\mathcal{L}_D$ according to eq. (22);
  Update the model weights (fixed and learnable as Fig. 2)(b);
  Training Feature Extraction Network: compute overall loss $\mathcal{L}_{Total}$ according to eq. (25);
  Update the model weights (fixed and learnable as Fig. 2)(c);

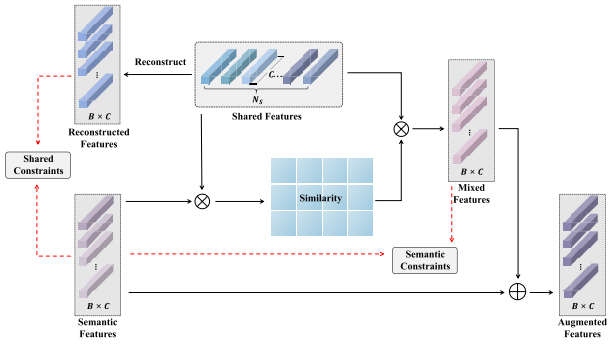


Fig. 5. Specific details of LCSFs.

1) Chikusei: This dataset consists of hyperspectral images captured in Ibaraki, Chikusai, Japan, using the Hyperspec-VNIR-CIRIS spectrometer. It features a ground sampling distance of 2.5 m, with a scene size of 2517 × 2335 pixels, and includes 128 spectral bands covering a wavelength range of 363–1018 nm. The dataset contains 19 categories, such as water and row crops. In the training process, we only select the sample number more than 1000 as training, as shown in Table I, we bold the training categories. The images and their associated ground-truth map are displayed in Fig. 6.

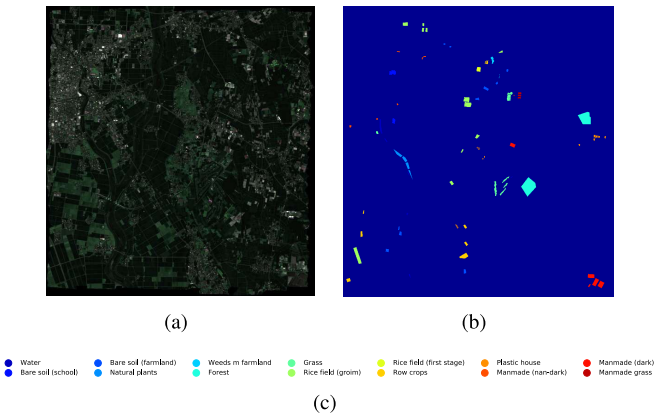


Fig. 6. Chikusei dataset. (a) False color image. (b) Ground-truth map. (c) Legend.

TABLE I
NUMBER OF SOURCE SAMPLES IN CHIKUSEI DATASET

Class No.	Class Name	Number of Samples
1	**Water**	2345
2	**Bare soil (school)**	2859
3	Bare soil (park)	236
4	**Bare soil (farmland)**	48525
5	**Natural plants**	4297
6	**Weeds m farmland**	1108
7	**Forest**	20516
8	**Grass**	6515
9	**Rice field (groim)**	13369
10	**Rice field (first stage)**	1268
11	**Row crops**	5961
12	**Plastic house**	2193
13	**Manmade (nan-dark)**	1220
14	**Manmade (dark)**	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	**Manmade grass**	1040
18	Asphalt	801
19	Paved ground	145
-	Total	77592

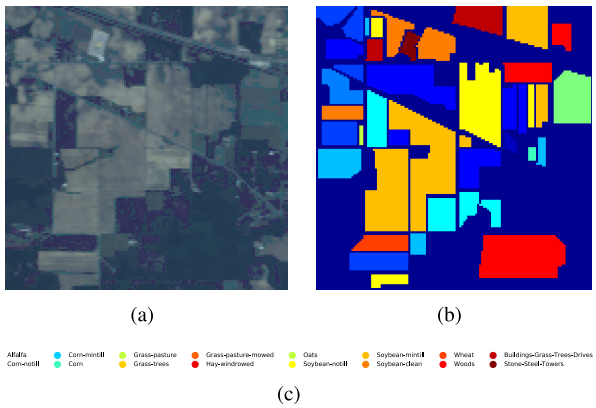


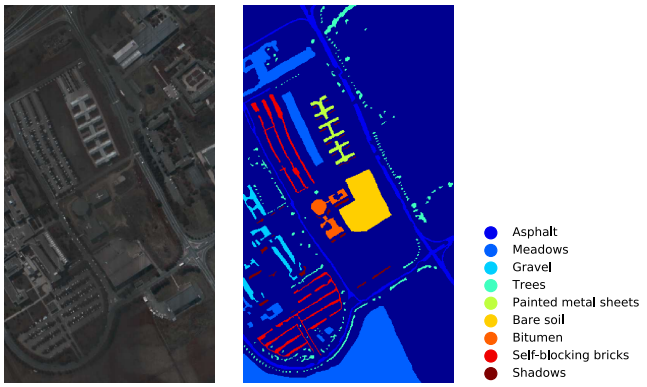
Fig. 7. IP dataset. (a) False color image. (b) Ground-truth map. (c) Legend.

2) Indian Pines: This dataset was acquired in 1992 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over agricultural fields in Indiana, USA. The dataset contains a 145 × 145 pixel spectral image, with each pixel point containing information on a continuous spectral band from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SUMMARY OF TRAINING AND TESTING SAMPLES IN IP DATASET

Class No.	Class Name	Number of Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	20
8	Hay-windrowed	473
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	336
16	Stone-Steel-Towers	93
-	Total	10249



(a) (b) (c)

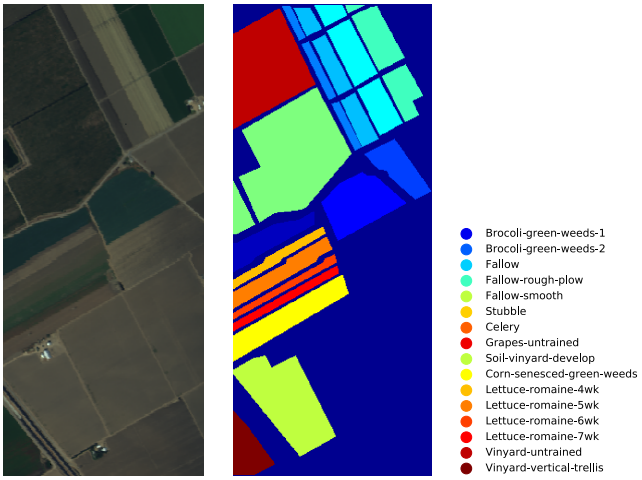
Fig. 8. PU dataset. (a) False color image. (b) Ground-truth map. (c) Legend.

TABLE III
SUMMARY OF TRAINING AND TESTING SAMPLES IN PU DATASET

Class No.	Class Name	Number of Samples
1	Asphalt	6631
2	Meadows	18549
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
-	Total	42776

0.4 to 2.5 μm, covering a total of 224 spectral bands. Since some bands cannot be reflected by water, 200 bands are generally used for practical studies. The dataset includes 16 categories, such as corn and wheat. The images and their associated ground-truth map are displayed in Fig. 7, and the categories are named and numbered, as shown in Table II.

3) University of Pavia: This dataset represents a portion of the hyperspectral imagery collected over the city of Pavia, Italy, by German Airborne Reflective Optics Spectrographic Imaging System (ROSIS-03) in 2003. The imaging system initially captured 115 spectral bands, out of which 103 bands were retained after removing noisy bands. The dataset includes



(a) (b) (c)

Fig. 9. Salinas hyperspectra dataset. (a) False color image. (b) Ground-truth map. (c) Legend.

TABLE IV
SUMMARY OF TRAINING AND TESTING SAMPLES IN SALINAS HYPERSPECTRAL DATASET

Class No.	Class Name	Number of Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3958
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1063
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
-	Total	54129

nine distinct feature classes, such as Trees, Asphalt, and Meadows, among others. The images and their associated ground-truth map are displayed in Fig. 8 and the categories are named and numbered, as shown in Table III.

4) Salinas: This dataset was collected over the Salinas Valley (SV), CA, USA, using the AVIRIS Imaging Spectrometer, with a spatial resolution of 3.7 m. Originally consisting of 224 spectral bands, 204 bands were retained after noise removal. The image dimensions are 512 $\times$ 217 pixels, with the pixels classified into 16 different categories, such as Fallow, Celery, and others. The images and their associated ground-truth map are displayed in Fig. 9, and the categories are named and numbered as shown in Table IV.

5) Houston: This dataset was obtained using the ITRES CASI-1500 sensor as part of the 2013 IEEE GRSS Data Fusion Competition. The dataset dimensions are 349 $\times$ 1905 pixels, encompassing 144 spectral bands ranging from 364 to 1046 nm. It includes 15 different classes, such as trees, soil, water, and more. The images and their associated ground-truth

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

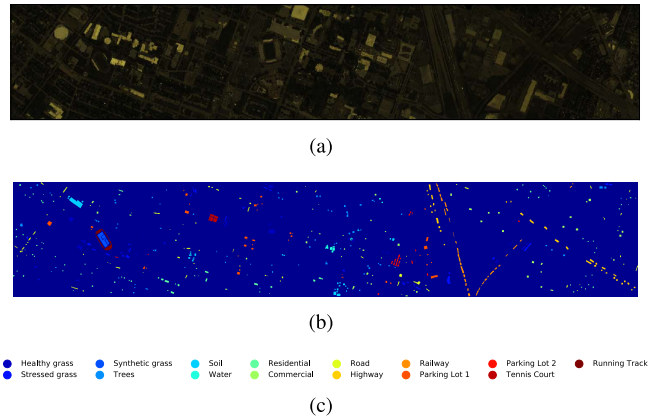


Fig. 10. University of HO dataset. (a) False color image. (b) Ground-truth map. (c) Legend.

TABLE V
SUMMARY OF TRAINING AND TESTING SAMPLES IN UNIVERSITY OF HO DATASET

Class No.	Class Name	Number of Samples
1	Healthy grass	1251
2	Stressed grass	1254
3	Synthetic grass	697
4	Trees	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1253
10	Highway	1227
11	Railway	1235
12	Parking Lot 1	1233
13	Parking Lot 2	469
14	Tennis Court	428
15	Running Track	660
-	TOTAL	150293

map are displayed in Fig. 10, and the categories are named and numbered as shown in Table V.

B. Experimental Setting

MSS-CFSL was developed using Python and the PyTorch framework. All experiments, including the comparative studies, were conducted on the same platform equipped with a GTX-3090 GPU. In MSS-CFSL, the input cube size was set to 9×9. To ensure fair comparisons, the EMAP preprocessing parameters used were consistent with those in CFSL-KT [47]. The model training utilized the Adam optimizer with a learning rate of 0.001 and a weight decay of $1e^{-4}$ to improve training stability. Training involved a total of 2000 iterations. For the SD, 200 labeled samples per class were randomly chosen to extract discriminative features, while for the TD, five labeled samples per class were used. In the comparative analysis, performance metrics such as overall accuracy (OA), average accuracy (AA), and the kappa coefficient were employed. To reduce bias and ensure fair evaluation, each experiment was repeated independently ten times.

C. Comparison of State-of-the-Art Methods

To validate the effectiveness of our proposed method, we compared it with 13 state-of-the-art approaches. Specifically, SVM [50], 3D-CNN [51], SSRN [45], and HybridSN [52] are supervised methods based on machine learning or deep learning algorithms utilizing CNN or Transformer architectures. In addition, we compared nine DA and FSL algorithms, including CMT [53], DCFSL [54], CMFSL [32], GiaCFSL [33], CFSL-KT [47], FDFSL [55], GCC-FSL [56], MLPA [57], and PR-CFSL [58]. All implementations were derived from the code repositories provided in the original papers, with hyperparameters set according to the optimal values reported in those studies. For supervised methods, training and testing were confined to the TD, with five samples per class randomly selected for training and the remaining samples used for testing. In cross-domain few-shot algorithms, 200 samples per class were randomly selected from SD to learn transferable class-discriminative features, while five samples per class from the TD were used for training, with the remaining samples reserved for testing. All reported results were obtained under the same conditions, using the best settings, and averaged over ten experimental runs on the same platform.

1) Classification Accuracy: Tables VI–IX present the OA, AA, and kappa coefficient results of ten comparative methods and our proposed MSS-CFSL on four different TD datasets: IP, UP, Salinas, and HO. Based on the results in Tables VI–IX, we can draw the following conclusions.

1) Machine learning and deep learning algorithms based on CNN or Transformer architectures, such as SVM [50], SSRN [45], 3D-CNN [51], HybridSN [52], and CMT [53], exhibit a consistent trend of inferior performance across all four datasets compared to other methods. This limitation primarily stems from two factors: 1) the extremely limited number of samples available for learning (only five samples per class used in Tables VI–IX), which leads to severe overfitting and 2) the absence of transferable knowledge learning mechanisms, which critically hampers their effectiveness in cross-domain scenarios. These observations highlight the fundamental challenges of applying purely supervised approaches to data-scarce cross-domain classification tasks.
2) DCFSL [54] pioneers the integration of FSL and DA by employing a conditional adversarial DA strategy to mitigate domain shift and learn transferable knowledge. Compared WITH SSRN, it achieves 1%–4% improvements across all four datasets, demonstrating that introducing transferable knowledge from SD can significantly enhance the learning of class-specific features in TD under limited sample availability. Subsequent approaches, including CMFSL [32], GiaCFSL [33], CFSL-KT [47], FDFSL [55], GCC-FSL [56], MLPA [57], PR-CFSL [58], and our proposed MSS-CFSL, further advance the field by designing sophisticated DA and alignment modules. These innovations lead to substantial accuracy improvements across all datasets, with our MSS-CFSL achieving state-of-the-art performance through its novel multiscale spectral–spatial feature fusion mechanism.
3) The meta-task simulates a sample-less learning scenario by randomly extracting a small number of samples from a large-scale source dataset. This training mechanism

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

CLASSIFICATION RESULTS OF METHODS ON THE IP DATASET

Class	Train	Test	SVM [50]	3D-CNN [51]	SSRN [45]	HybridSN [52]	CMT [53]	DCFSL [54]	CMFSL [32]	GiaCFSL [33]	CFSL-KT [47]	FDFSL [55]	GCC-FSL [56]	MLPA [57]	PR-CFSL [58]	MSS-CFSL
1	5	41	78.78	81.67	96.10	54.17	97.32	93.17	99.02	91.70	**99.76**	98.28	87.46	98.54	97.56	**99.76**
2	5	1423	35.46	24.99	40.55	50.38	**61.49**	40.26	45.47	41.01	52.83	51.65	40.44	57.08	58.03	55.54
3	5	825	41.55	26.92	56.62	50.46	46.96	50.96	57.16	45.96	69.47	69.32	47.22	60.27	66.47	**70.48**
4	5	232	58.27	44.80	68.59	30.45	85.69	76.21	80.30	79.91	85.55	**91.42**	79.01	83.15	89.22	90.48
5	5	478	66.33	68.75	77.40	47.47	68.20	73.26	77.28	68.14	81.58	72.55	66.74	75.96	81.63	**83.01**
6	5	725	78.78	82.08	95.90	71.16	**98.34**	84.84	84.32	77.82	93.03	84.72	80.65	86.95	92.73	96.75
7	5	23	91.30	94.43	99.44	45.18	**100.0**	99.13	99.13	98.70	98.84	**99.57**	97.83	99.13	**100.0**	99.57
8	5	473	71.22	62.26	74.16	72.77	**100.0**	83.97	87.34	84.71	97.80	84.46	88.35	90.70	85.29	**99.59**
9	5	15	86.66	99.00	**100.0**	14.60	**100.0**	**100.0**	**100.0**	98.67	99.98	**100.0**	99.33	98.67	**100.0**	**100.0**
10	5	967	39.97	54.22	56.40	45.22	81.44	61.28	65.58	58.75	**81.58**	63.65	61.96	68.49	64.47	76.25
11	5	2450	34.93	41.46	48.53	67.01	71.44	60.71	57.27	60.84	74.03	63.08	58.70	59.84	67.65	**79.56**
12	5	588	25.68	30.34	47.29	47.87	**77.35**	47.02	49.46	38.16	54.80	59.18	46.00	55.58	59.97	67.39
13	5	200	91.10	96.41	98.10	55.88	99.15	96.95	99.30	98.20	95.20	99.30	**98.55**	97.20	**99.90**	99.60
14	5	1260	58.33	63.67	79.55	67.42	**92.44**	83.72	85.54	77.14	92.13	87.11	77.71	86.63	91.93	89.54
15	5	381	27.95	50.47	76.06	53.41	**97.35**	68.61	75.56	62.05	**91.96**	77.74	64.67	78.45	78.24	91.20
16	5	88	88.63	87.35	97.83	42.93	**100.0**	95.68	99.20	98.52	97.55	97.95	99.09	98.30	99.66	**100.0**
OA(%)	-	-	46.89 ± 3.46	48.57 ± 2.76	61.68 ± 3.61	53.03 ± 4.11	77.34 ± 2.57	64.51 ± 2.60	66.69 ± 3.12	61.84 ± 4.30	76.42 ± 6.06	70.26 ± 1.49	62.55 ± 2.60	70.11 ± 2.62	73.78 ± 1.80	**79.42 ± 3.11**
AA(%)	-	-	60.93 ± 2.91	63.03 ± 2.49	75.78 ± 1.97	68.48 ± 1.85	86.07 ± 2.16	75.98 ± 1.94	78.87 ± 0.86	73.77 ± 2.50	85.48 ± 2.75	81.25 ± 1.09	74.61 ± 1.81	80.93 ± 1.58	83.30 ± 1.95	**87.43 ± 1.60**
Kappa	-	-	40.86 ± 3.63	42.52 ± 2.77	57.07 ± 3.79	47.89 ± 4.14	74.56 ± 1.58	60.03 ± 2.86	62.32 ± 2.87	57.00 ± 4.46	73.42 ± 6.54	66.61 ± 1.61	57.91 ± 2.75	66.41 ± 2.76	70.34 ± 2.01	**76.79 ± 3.42**

TABLE VII

CLASSIFICATION RESULTS OF METHODS ON THE PU DATASET

Class	Train	Test	SVM [50]	3D-CNN [51]	SSRN [45]	HybridSN [52]	CMT [53]	DCFSL [54]	CMFSL [32]	GiaCFSL [33]	CFSL-KT [47]	FDFSL [55]	GCC-FSL [56]	MLPA [57]	PR-CFSL [58]	MSS-CFSL
1	5	6626	62.76	63.52	81.12	75.26	76.94	81.26	77.62	77.89	**98.89**	80.51	77.80	80.33	88.65	96.90
2	5	18544	58.62	64.18	66.51	91.67	**93.70**	81.34	81.44	77.26	90.69	77.42	78.61	85.50	85.26	93.47
3	5	2094	48.65	40.16	52.91	48.88	**81.80**	53.65	54.35	56.74	77.21	58.01	52.81	65.01	61.89	80.11
4	5	3095	82.37	94.34	85.21	59.36	78.42	82.80	**95.00**	88.38	92.22	91.12	89.44	93.24	93.25	93.49
5	5	1340	90.02	97.18	98.17	84.38	**100.0**	96.60	97.86	96.10	99.72	97.90	96.31	99.07	99.79	**100.0**
6	5	5024	57.54	45.80	71.81	78.67	64.24	69.45	68.37	70.81	93.65	73.44	65.57	73.74	71.79	**95.03**
7	5	1325	92.55	91.62	96.82	69.38	96.27	88.28	85.74	84.32	99.50	93.91	84.95	78.09	81.06	**99.72**
8	5	3680	50.14	52.15	83.91	54.35	58.58	62.41	67.68	64.68	90.22	72.73	67.63	68.62	78.29	**92.22**
9	5	942	99.83	99.85	**99.86**	48.66	98.16	98.50	99.14	97.31	96.57	94.62	97.06	99.25	97.25	93.31
OA(%)	-	-	62.56 ± 4.35	64.55 ± 5.44	74.23 ± 4.95	74.33 ± 3.53	84.18 ± 3.25	78.84 ± 3.58	79.65 ± 4.21	76.54 ± 5.62	92.40 ± 2.59	78.59 ± 3.82	76.68 ± 4.12	81.90 ± 3.45	83.26 ± 1.84	**93.81 ± 1.92**
AA(%)	-	-	71.38 ± 2.35	72.14 ± 3.33	81.86 ± 3.02	73.94 ± 2.26	84.37 ± 4.51	80.48 ± 1.24	80.80 ± 2.39	79.25 ± 2.49	93.19 ± 7.35	82.18 ± 2.74	78.91 ± 1.24	82.53 ± 1.72	84.13 ± 1.87	**93.77 ± 1.58**
Kappa	-	-	53.67 ± 4.51	55.79 ± 5.94	67.76 ± 5.52	66.88 ± 4.22	78.06 ± 2.73	72.78 ± 4.05	72.78 ± 4.30	70.16 ± 6.41	90.11 ± 3.30	72.63 ± 4.46	70.15 ± 4.52	76.57 ± 4.09	78.72 ± 2.07	**91.88 ± 2.46**

TABLE VIII

CLASSIFICATION RESULTS OF METHODS ON THE SALINAS DATASET

Class	Train	Test	SVM [50]	3D-CNN [51]	SSRN [45]	HybridSN [52]	CMT [53]	DCFSL [54]	CMFSL [32]	GiaCFSL [33]	CFSL-KT [47]	FDFSL [55]	GCC-FSL [56]	MLPA [57]	PR-CFSL [58]	MSS-CFSL
1	5	2004	98.08	90.35	98.53	94.71	**100.0**	**99.55**	99.50	98.53	97.28	99.09	99.31	99.04	**99.96**	97.57
2	5	3721	93.60	91.58	99.02	97.07	99.32	99.49	99.70	98.99	99.31	99.25	99.14	98.44	**99.85**	99.25
3	5	1971	84.31	57.05	94.89	95.36	**99.78**	92.80	90.28	87.77	93.81	89.74	85.26	94.62	95.59	95.59
4	5	1389	99.19	97.81	96.28	78.67	97.95	99.55	99.38	98.24	99.83	99.34	98.75	97.05	99.23	**99.86**
5	5	2673	90.77	89.62	95.57	89.90	**96.17**	92.67	93.21	88.75	90.65	90.98	87.15	95.72	94.09	92.43
6	5	3954	98.23	98.56	99.21	88.89	**100.0**	99.21	99.38	98.71	99.39	98.76	98.91	99.28	**99.60**	99.24
7	5	3574	96.73	99.34	99.50	98.40	**100.0**	98.88	99.19	98.12	**100.0**	98.89	98.04	96.89	99.84	**100.0**
8	5	11266	60.00	49.23	49.92	86.21	82.42	74.21	69.96	74.05	83.67	80.47	73.06	70.23	82.49	**88.73**
9	5	6198	94.11	93.29	98.17	98.74	99.81	99.30	99.75	97.50	99.62	98.93	97.67	97.15	**99.83**	99.50
10	5	3273	73.02	60.86	88.28	90.88	93.17	85.95	84.25	78.25	95.35	81.98	79.62	82.32	87.12	**95.52**
11	5	1063	87.60	81.72	96.09	91.10	**100.0**	97.88	98.79	95.10	99.26	95.16	94.33	93.41	98.86	98.03
12	5	1922	96.24	98.56	**99.90**	96.45	96.83	99.41	99.60	97.55	96.56	99.24	99.39	96.68	99.69	96.59
13	5	911	97.99	97.61	**99.67**	89.75	90.31	99.15	99.36	98.16	85.22	98.63	97.56	99.05	99.55	95.99
14	5	1065	90.28	93.52	94.13	94.70	95.41	98.27	98.75	97.77	99.21	98.27	97.04	98.69	98.70	**99.36**
15	5	7263	57.88	64.88	80.71	77.31	79.53	75.19	76.78	73.49	**92.41**	83.15	74.18	72.06	79.72	88.03
16	5	1802	83.07	87.34	94.34	94.78	98.81	90.65	90.80	86.70	**100.0**	89.85	86.53	93.36	96.17	**100.0**
OA(%)	-	-	80.62 ± 2.50	77.32 ± 2.22	84.45 ± 2.52	90.41 ± 1.40	91.78 ± 2.47	89.12 ± 1.56	88.41 ± 2.27	87.33 ± 2.17	93.97 ± 2.26	90.85 ± 1.36	87.23 ± 1.84	87.25 ± 1.35	92.03 ± 0.83	**94.56 ± 0.94**
AA(%)	-	-	87.57 ± 1.04	84.46 ± 1.59	92.26 ± 1.33	92.46 ± 3.80	95.29 ± 3.12	93.89 ± 0.80	93.67 ± 1.18	91.73 ± 1.53	95.74 ± 2.48	93.86 ± 1.35	91.62 ± 1.40	82.75 ± 0.26	95.61 ± 0.77	**95.95 ± 1.80**
Kappa	-	-	78.50 ± 2.66	74.93 ± 2.36	82.80 ± 2.73	89.35 ± 1.55	91.13 ± 2.83	87.91 ± 1.70	87.14 ± 2.48	85.94 ± 2.39	93.30 ± 2.47	89.82 ± 1.49	85.83 ± 2.00	85.85 ± 1.45	91.13 ± 0.92	**93.94 ± 1.05**

TABLE IX

CLASSIFICATION RESULTS OF METHODS ON THE HO DATASET

Class	Train	Test	SVM [50]	3D-CNN [51]	SSRN [45]	HybridSN [52]	CMT [53]	DCFSL [54]	CMFSL [32]	GiaCFSL [33]	CFSL-KT [47]	FDFSL [55]	GCC-FSL [56]	MLPA [57]	PR-CFSL [58]	MSS-CFSL
1	5	1246	87.35	87.56	**91.28**	72.72	79.21	88.80	88.88	88.44	86.55	87.14	87.37	90.70	84.48	85.12
2	5	1249	90.17	81.22	91.07	72.47	79.18	86.89	91.43	89.34	69.46	**91.87**	89.14	80.46	90.52	84.78
3	5	692	95.93	92.83	99.40	90.60	**100.0**	98.95	95.69	95.23	99.22	95.88	95.92	95.34	94.06	97.81
4	5	1239	90.43	91.82	95.61	61.36	**98.17**	95.42	95.50	94.05	86.23	92.11	95.17	91.83	96.96	85.90
5	5	1237	88.01	82.89	93.48	83.30	**100.0**	95.05	95.04	90.97	93.51	97.58	90.95	95.98	96.43	96.32
6	5	320	71.81	85.52	85.68	68.03	87.53	80.69	85.19	76.44	92.50	83.06	79.47	86.14	75.45	**94.84**
7	5	1263	50.83	66.62	75.55	57.49	77.00	74.16	78.54	70.84	71.55	79.22	69.18	67.87	**83.45**	77.28
8	5	1239	26.25	48.02	42.66	**69.33**	49.77	48.53	52.63	47.68	52.62	50.06	46.63	44.54	53.95	53.95
9	5	1247	68.35	61.93	64.76	61.82	72.19	68.50	74.80	63.34	70.58	75.16	63.71	58.10	64.02	**76.34**
10	5	1222	37.41	40.59	76.18	60.81	**94.57**	73.35	66.78	67.35	86.47	68.90	63.53	73.54	61.54	87.52
11	5	1230	50.75	66.25	72.85	73.88	85.65	64.96	69.54	64.92	84.41	73.37	66.41	57.08	67.60	**87.60**
12	5	1228	20.93	29.18	54.37	66.76	59.63	72.44	68.36	64.15	67.13	55.11	63.07	**73.40**	59.75	67.41
13	5	464	20.08	83.35	78.62	52.54	80.12	85.73	94.72	88.34	60.84	91.47	88.77	**92.57**	81.40	84.59
14	5	423	88.34	95.59	98.75	91.54	**100.0**	91.49	92.41	88.46	99.95	96.50	88.06	88.27	81.86	**100.0**
15	5	655	95.90	98.79	99.07	74.45	99.51	95.51	97.21	89.76	99.45	95.44	90.24	94.79	94.19	**100.0**
OA(%)	-	-	63.99 ± 2.02	70.20 ± 3.86	78.86 ± 3.04	66.78 ± 2.94	82.07 ± 2.18	79.44 ± 1.83	80.90 ± 2.96	76.69 ± 2.00	79.35 ± 1.92	79.90 ± 2.20	76.32 ± 1.80	76.21 ± 2.46	77.64 ± 1.47	**82.73 ± 1.26**
AA(%)	-	-	65.50 ± 1.89	74.15 ± 3.24	81.29 ± 2.48	70.04 ± 2.19	84.29 ± 1.79	81.37 ± 1.70	83.11 ± 2.36	78.62 ± 1.65	81.36 ± 2.51	82.19 ± 1.98	78.51 ± 1.77	79.37 ± 2.25	79.04 ± 1.38	**85.06 ± 0.76**
Kappa	-	-	61.12 ± 2.16	67.83 ± 4.15	77.17 ± 3.29	64.17 ± 3.15	80.70 ± 2.37	77.79 ± 1.97	79.37 ± 3.19	74.83 ± 2.14	77.68 ± 2.09	78.28 ± 2.38	74.44 ± 1.92	74.32 ± 2.66	75.54 ± 1.59	**81.21 ± 1.22**

enables the model to efficiently extract key information under limited sample conditions, thus significantly improving its generalization ability. Experiments show that DCFSL, CMFSL, GiaCFSL, CFSL-KT, FDFSL, GCC-CFSL, MLPA, PR-CFSL, and our proposed MSS-CFSL fully utilize the idea of meta-tasks. By training in meta-tasks, these models exhibit enhanced transferability and discriminative ability. Especially in the case of data scarcity, the method substantially improves the performance of FSL.

4) Based on existing studies, our proposed MSS-CFSL significantly improves the performance of cross-domain few-sample HSIC by effectively learning low-level stylistic features and high-level shared semantic features, and fusing spectral and spatial information. The experimental results show that the OA of MSS-CFSL is improved by 17.74%, 19.58%, 10.11%, and 3.84% compared with SSRN on IP, UP, Salinas, and HO datasets, respectively. Compared with the traditional FSL and multiple cross-domain methods (e.g., DCFSL, CMFSL,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

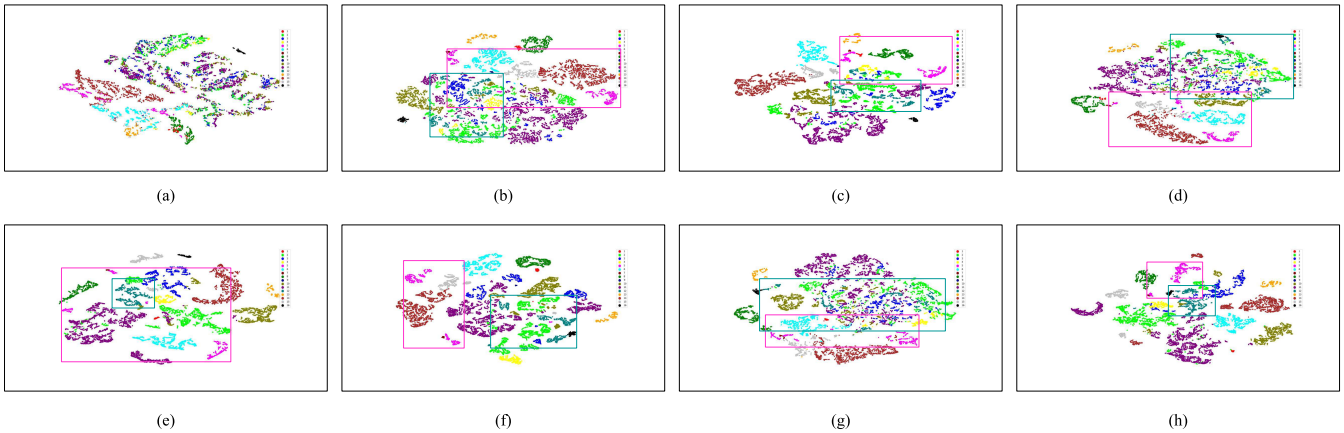


Fig. 11. Feature visualization of results from various methods on the TD IP dataset including (a) original feature distribution, (b) DCFSL, (c) CMFSL, (d) GiaCFSL, (e) CFSL-KT, (f) FDFSL, (g) GCC-FSL, and (h) MSS-CFSL.

TABLE X
PARAMETERS, FLOPS, AND COMPUTATION TIME (SECONDS) ON EACH DATASET WITH DIFFERENT METHODS

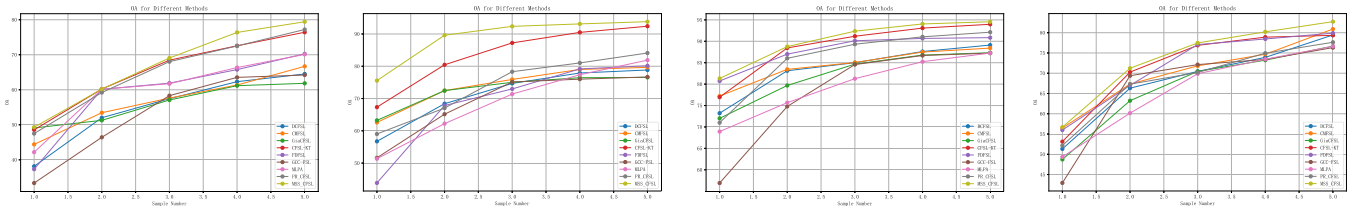
	Methods	DCFSL	CMFSL	GiaCFSL	CFSL-KT	FDFSL	GCC-FSL	PR-CFSL	MLPA	MSS-CFSL
IP	Params	**0.050496M**	0.626976M	0.438995M	0.724816M	0.19055M	0.205392M	4.261721M	12.666452M	0.835376M
	FLOPs	42.58168M	28.199728M	8868.393168M	**12.144768M**	52.722785M	43.312848M	47.3203M	3412.614424M	50.1128M
	Training Time	436.61	**364.13**	1037.01	417.15	491.95	1376.98	1461.56	2202.40	1092
	Testing Time	1.88972	0.73	1.85	**0.48**	2.86	1.83	1.47	1.01	1.15
UP	Params	**0.047996M**	0.626976M	0.438995M	0.724816M	0.18085M	0.195692M	4.257831M	12.6534784M	0.833577M
	FLOPs	42.379180M	28.199728M	8868.393168M	**12.144768M**	51.937085M	42.527148M	46.587123M	3412.614424M	50.111008M
	Training Time	233.22	**147.97**	591.12	398.32	310.53	720.84	917.27	1849.13	1039
	Testing Time	6.83	2.56	6.79	**2.1**	16.23	6.85	5.46	7.65	4.97
Salinas	Params	**0.058096M**	0.626976M	0.438995M	0.724816M	0.19095M	0.205792M	4.267314M	12.666452M	0.835376M
	FLOPs	43.197280M	28.199728M	8868.393168M	**12.144768M**	52.755185M	43.345248M	47.390371M	3412.614424M	50.1128M
	Training Time	398.01	**316.49**	965.85	440.45	534.4	1318.25	1516.89	3041.57	1043
	Testing Time	10.18	4.04	10.38	**2.74**	24.05	10.42	7.95	6.64	
Houston	Params	**0.050335M**	0.626976M	0.438995M	0.724816M	0.18495M	0.199792M	4.267326M	12.6534784M	0.835119M
	FLOPs	42.58152M	28.199728M	8868.393168M	**12.144768M**	52.269185M	42.859248M	47.315624M	3412.614424M	50.112544M
	Training Time	312.77	**229.86**	791.58	424.38	370.33	1094.18	1578.36	3347.71	1051
	Testing Time	2.46	0.95	2.48	**0.75**	4.03	2.49	5.78	2.59	1.74

GiaCFSL, CFSL-KT, FDFSL, GCC-CFSL, MLPA, and PR-CFSL), MSS-CFSL improves the average OA on the four datasets by 11.15%, 12.82%, 5.04%, and 4.42%, respectively. In addition, compared with the best-performing method, CFSL-KT, MSS-CFSL improves OA by 3.00%, 1.41%, 0.59%, and 3.38% and has a smaller standard deviation, indicating that it demonstrates higher stability in the random sampling case. These results validate that MSS-CFSL significantly improves the accuracy and robustness of the model in cross-domain few-sample hyperspectral classification tasks through the strategy of multilevel feature fusion and enhancement.

2) Feature Visualization: In order to visually compare the feature representations of different methods, we use t-distributed stochastic neighbor embedding (t-SNE) [59] to project the features (see Fig. 11). This figure shows the feature distribution of MSS-CFSL alongside several other FSL methods on the IP dataset, comparing it with the original feature distribution of the dataset. It can be observed that the original feature distribution is more chaotic, with category differences being difficult to distinguish. In contrast, MSS-CFSL forms more distinct category clusters in the t-SNE embedding space, exhibiting significantly higher intraclass compactness and greater interclass separation. This indicates that by enhancing low-level stylistic features and high-level semantic features, MSS-CFSL is more effective in capturing discriminative information in cross-domain tasks. In comparison, other methods show a more scattered feature distribution with unclear interclass boundaries, making it challenging to differentiate between classes. These results suggest that MSS-CFSL demonstrates superior feature differentiation when addressing domain shift and few-shot problems, leading to a significant improvement in classification accuracy.

3) Time Complexity: In terms of time performance, our proposed MSS-CFSL demonstrates significant advantages, particularly in training and testing times compared to other competitive methods (Table X). For example, on the IP dataset, the training time for MSS-CFSL is 1092 s, which, although higher than CMFSL's 364.13 s, shows greater stability across other datasets, especially in complex datasets such as HO. Regarding the number of parameters and floating point operations per second (FLOPs), MSS-CFSL strikes a good balance in performance. Specifically, the number of parameters for MSS-CFSL is consistently 0.835376M across various datasets (both IP and Salinas), which, although slightly higher than CMFSL (0.626976M) and DCFSL (0.050496M), aims to more effectively capture features to enhance classification performance. In terms of FLOPs, MSS-CFSL's computational

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

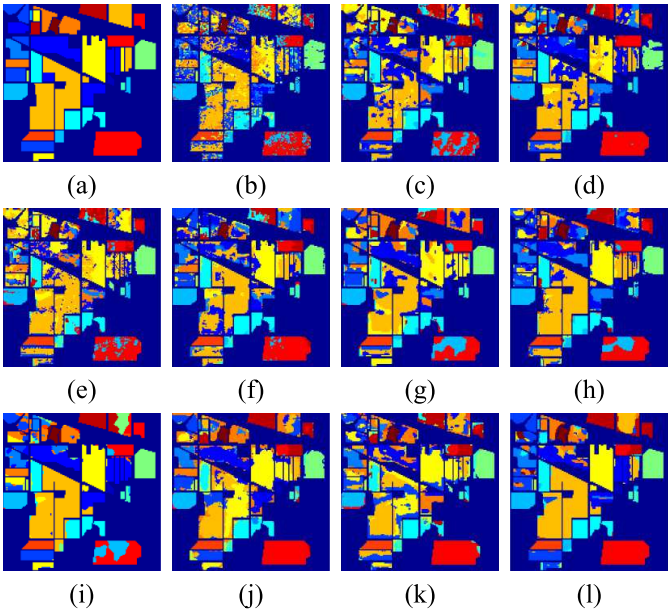
Fig. 12. Different numbers of labeled samples from TD. (a) IP. (b) UP. (c) Salinas. (d) HO.

complexity is 50.1128M, slightly exceeding that of DCFSL (42.58168M) and UP (50.111008M); yet, it remains within an acceptable range. While FLOPs are relatively high, this complexity is justifiable in improving model accuracy and feature learning capability. In summary, the design of MSS-CFSL regarding the number of parameters and FLOPs reflects a balance between performance and efficiency, ensuring that it maintains a high classification accuracy while still possessing acceptable computational complexity, making it suitable for deployment in practical applications.

4) Different Numbers of Labeled Samples From TD: As shown in Fig. 12, the analysis across four datasets highlights the significant advantages of MSS-CFSL, particularly in few-shot scenarios. The model adaptively combines low-level style features with high-level shared features, maintaining robust accuracy even as sample size increases. For instance, on the IP dataset, MSS-CFSL achieves 79.42% OA with five samples, outperforming CFSL-KT (76.42%) and other baseline methods. A similar trend is observed on the SV dataset, where MSS-CFSL reaches the highest OA of 94.56%, surpassing CFSL-KT (93.97%). These results demonstrate our method's effectiveness in cross-domain tasks through synergistic enhancement of low-level style features and utilization of high-level semantic information.

Notably, MSS-CFSL shows remarkable performance improvements under extreme data scarcity. On the Pavia University (PU) dataset, it achieves 92.36% OA with only three samples and 93.81% with five samples, exceeding all comparative methods. This robustness is further validated on the HO dataset, where MSS-CFSL attains the highest OA of 82.73%. The consistent superiority across varying domains and sample sizes confirms the model's capability to hierarchically integrate multiscale features (from spectral–textural details to semantic abstractions), effectively addressing the dual challenges of domain shift and sample insufficiency in few-shot hyperspectral classification.

5) Classification Map Visualization: Figs. 13–16 present visual comparisons of ten competing methods, our proposed MSS-CFSL, and the ground-truth maps. The visualization results demonstrate that MSS-CFSL achieves the closest alignment with the ground truth, particularly in preserving fine-grained spatial structures and suppressing classification noise. This visual superiority quantitatively corresponds to the performance metrics in Tables VI–IX, confirming the method's effectiveness and robustness in cross-domain few-shot classification.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 13. Visualization of results from various methods on TD IP dataset, including (a) ground-truth map (100%), (b) SVM (46.89%), (c) 3-D-CNN (48.57%), (d) SSRN (61.68%), (e) HybridSN (53.03%), (f) DCFSL (64.51%), (g) CMFSL (66.69%), (h) GiaCFSL (61.84%), (i) CFSL-KT (76.42%), (j) FDFSL (70.26%), (k) GCC-FSL (62.55%), and (l) MSS-CFSL (79.42%).

6) More Cross-Domain Experiments: To validate cross-source generalization, we conducted ten independent experiments with IP→UP migration. As shown in Table XI, MSS-CFSL significantly outperforms supervised baselines and the suboptimal CFSL-KT, robustly confirming its cross-source generalization capability.

D. Ablation Study

1) Ablation Comparison of Single Component: To investigate the contributions of individual modules in MSS-CFSL, we conducted an ablation study focusing on three core components: SADA_V2 (spatial adaptation), SADA_V3 (spectral–spatial adaptation), and LCSF (learnable class-shared features), as detailed in Table XII. We systematically removed each module while maintaining other components, creating three variants: 1) MSS-CFSL w/o SADA_V2; 2) MSS-CFSL w/o SADA_V3; and 3) MSS-CFSL w/o LCSF. The complete MSS-CFSL achieves optimal performance across all datasets (bold values in Table XII), with any module removal causing performance degradation, confirming their synergistic importance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

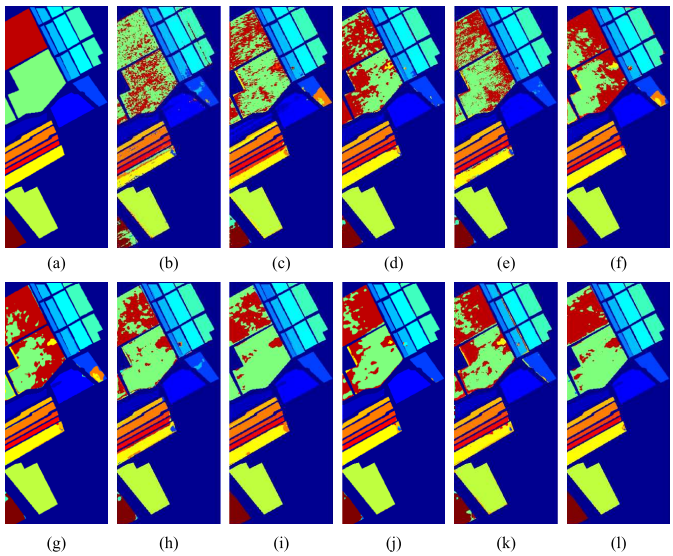


Fig. 14. Visualization of results from various methods on the TD salians dataset including (a) ground-truth map (100%), (b) SVM (80.62%), (c) 3-D-CNN (77.32%), (d) SSRN (84.45%), (e) HybridSN (90.41%), (f) DCFSL (89.12%), (g) CMFSL (88.41%), (h) GiaCFSL (87.33%), (i) CFSL-KT (93.97%), (j) FDFSL (90.85%), (k) GCC-FSL (87.32%), and (l) MSS-CFSL (94.56%).

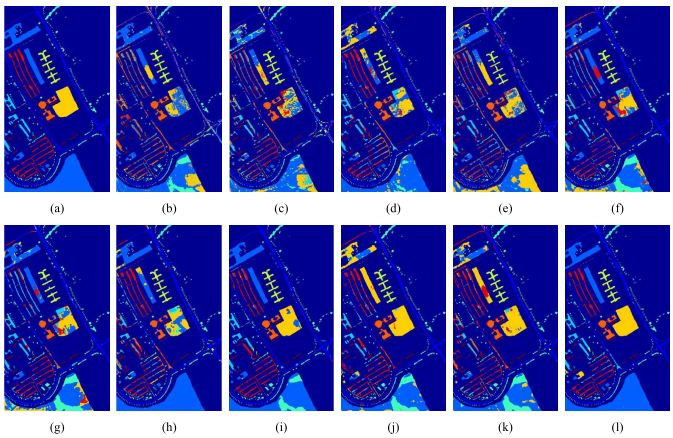


Fig. 15. Visualization of results from various methods on TD PU dataset including (a) ground-truth map (100%), (b) SVM (62.56%), (c) 3-D-CNN (64.55%), (d) SSRN (74.23%), (e) HybridSN (74.33%), (f) DCFSL (78.84%), (g) CMFSL (79.65%), (h) GiaCFSL (76.54%), (i) CFSL-KT (92.40%), (j) FDFSL (78.59%), (k) GCC-FSL (76.68%), and (l) MSS-CFSL (93.81%).

Specifically, eliminating SADA_V2 reduces OA by 0.6%–2.6% due to impaired spatial feature alignment through 2D-CNN. Removing SADA_V3 decreases accuracy by 0.4%–1.6% as it disrupts spectral–spatial joint adaptation via 3-D-CNN. The absence of LCSF causes a 0.5%–1.4% OA drop by weakening cross-domain semantic consistency. These results demonstrate that: 1) spatial–spectral decoupled adaptation (SADA_V2/V3) effectively addresses domain shifts and 2) LCSF enables robust cross-domain knowledge transfer.

2) Sequential Ablation Comparison: To further validate the effectiveness of MSS-CFSL, we conducted additional ablation studies by integrating our proposed modules (SADA_V2, SADA_V3, and LCSF) into CFSL-KT, the best-performing baseline method. As shown in Table XIII, the progressive addition of these modules yields consistent performance improvements across all four datasets, with OA increasing by 0.8%–2.5%, AA by 0.7%–3.0%, and kappa by 1.1%–2.2%. Specifically, adding SADA_V2 enhances spatial feature adaptation, contributing 0.4%–0.8% OA gains;

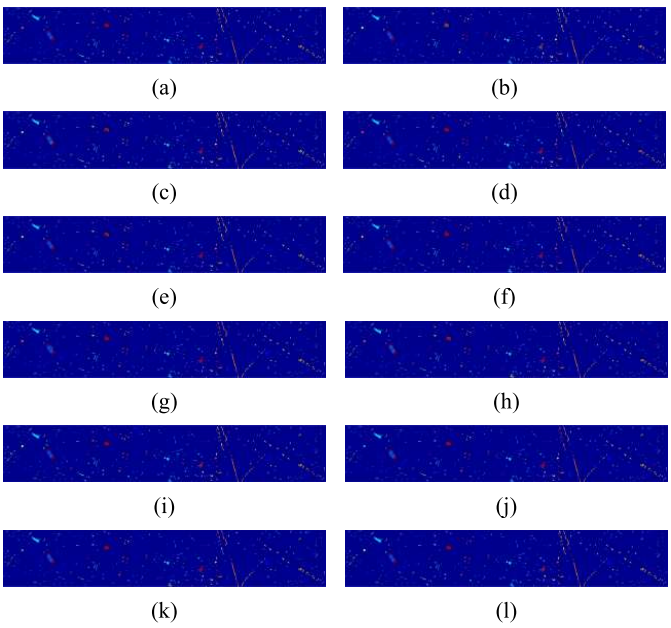


Fig. 16. Visualization of results from various methods on the TD HO dataset including (a) ground-truth map (100%), (b) SVM (63.99%), (c) 3-D-CNN (70.20%), (d) SSRN (78.86%), (e) HybridSN (66.78%), (f) DCFSL (79.44%), (g) CMFSL (80.90%), (h) GiaCFSL (76.69%), (i) CFSL-KT (79.35%), (j) FDFSL (79.90%), (k) GCC-FSL (76.32%), and (l) MSS-CFSL (82.73%).

TABLE XI

CLASSIFICATION RESULTS OF METHODS ON THE PU DATASET (SD: IP)

Class	Train	Test	SVM [50]	3D-CNN [51]	SSRN [45]	HybridSN [52]	CMT [53]	CFSL-KT [47]	MSS-CFSL
1	5	6626	62.76	63.52	81.12	75.26	76.94	97.61	**98.98**
2	5	18544	58.62	64.18	66.51	91.67	**93.70**	91.25	92.74
3	5	2094	48.65	40.16	52.91	48.88	**81.80**	76.89	77.28
4	5	3095	82.37	94.34	85.21	59.36	78.42	92.22	93.53
5	5	1340	90.02	97.18	98.17	84.38	**100.0**	99.49	**99.90**
6	5	5024	57.54	45.80	71.81	78.67	64.24	93.18	**98.27**
7	5	1325	92.55	91.62	96.82	69.38	96.27	**99.52**	99.17
8	5	3680	50.14	52.15	83.91	54.35	58.58	90.17	**90.49**
9	5	942	99.83	99.85	**99.86**	48.66	98.16	96.16	90.69
OA(%)	-	-	62.56 ± 4.35	64.55 ± 5.44	74.23 ± 4.95	74.33 ± 3.53	84.18 ± 3.25	92.48 ± 2.17	**93.84 ± 2.08**
AA(%)	-	-	71.38 ± 2.35	72.14 ± 3.33	81.86 ± 3.02	73.94 ± 2.26	84.37 ± 4.51	93.13± 1.25	**93.45 ± 0.92**
Kappa	-	-	53.67 ± 4.51	55.79 ± 5.94	67.76 ± 5.52	66.88 ± 4.22	78.06 ± 2.73	91.03 ± 1.75	**91.97 ± 2.64**

TABLE XII

ABLATION COMPARISON OF SINGLE COMPONENT ON THE FOUR DATASETS

Data set	Method	OA(%)	AA(%)	Kappa
IP	w/o SADA_V2	78.37±3.87	86.43±2.32	75.59±4.28
	w/o SADA_V3	78.41±3.72	86.27±2.41	75.65±4.12
	w/o LCSF	78.22±3.83	86.62±2.06	75.51±4.18
	MSS-CFSL	**79.42±3.11**	**87.43±1.60**	**76.79±3.42**
UP	w/o SADA_V2	93.36±2.45	93.24±1.46	91.35±3.14
	w/o SADA_V3	93.21±2.66	93.06±1.54	91.16±3.50
	w/o LCSF	92.96±2.76	92.71±1.99	90.81±3.52
	MSS-CFSL	**93.81±1.92**	**93.77±1.58**	**91.88±2.46**
Salinas	w/o SADA_V2	93.81±1.98	96.51±1.08	93.12±2.20
	w/o SADA_V3	94.11±1.90	96.10±1.57	93.45±2.11
	w/o LCSF	94.02±1.57	96.00±1.01	93.36±1.74
	MSS-CFSL	**94.56±0.94**	**95.95±1.80**	**93.94±1.05**
Houston	w/o SADA_V2	80.05±2.13	81.45±2.05	79.07±1.87
	w/o SADA_V3	81.17±1.68	83.07±1.89	80.50±2.01
	w/o LCSF	81.35±1.45	84.00±1.62	80.78±1.56
	MSS-CFSL	**82.73±1.26**	**85.06±0.76**	**81.21±1.22**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

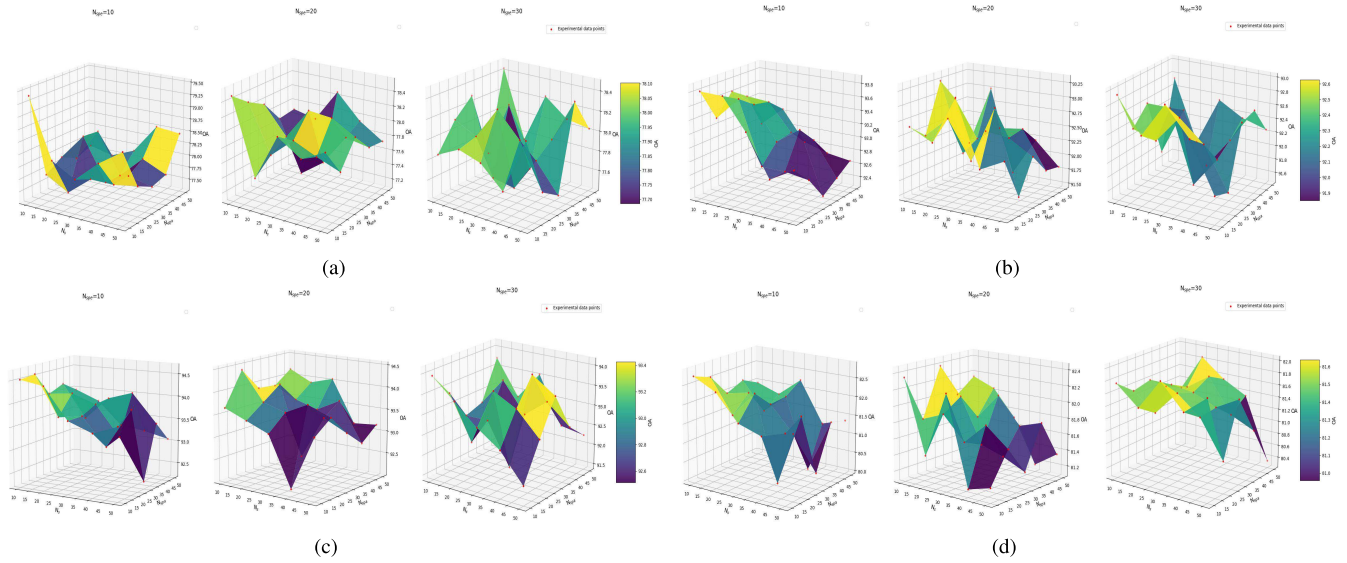


Fig. 17. Parameter analysis of N, N_{spa}, and N_{spe} on the four datasets. (a) IP. (b) UP. (c) Salinas. (d) HO.

TABLE XIII
SEQUENTIAL ABLATION COMPARISON ON THE FOUR DATASETS

Data set	SADA_V2	SADA_V3	LCSF	OA(%)	AA(%)	Kappa
IP	×	×	×	76.42±6.06	85.58±2.75	73.42±6.54
	✓	×	×	77.20±5.30	86.25±2.43	74.10±5.67
	✓	✓	×	77.75±4.35	86.80±2.20	74.60±4.37
	✓	✓	✓	**78.15±2.87**	**87.15±2.20**	**75.00±2.16**
UP	×	×	×	92.40±2.59	93.19±7.35	90.11±3.30
	✓	×	×	92.85±2.23	93.50±6.07	90.65±2.84
	✓	✓	×	93.15±2.51	93.75±5.24	91.05±2.43
	✓	✓	✓	**93.40±2.03**	**93.95±4.02**	**91.40±2.05**
Salinas	×	×	×	93.97±2.26	95.74±2.48	93.30±2.47
	✓	×	×	94.35±2.12	96.05±2.30	93.75±2.28
	✓	✓	×	94.65±2.20	96.30±2.15	94.10±2.10
	✓	✓	✓	**94.81±2.05**	**96.50±2.12**	**94.40±1.95**
Houston	×	×	×	79.35±1.92	81.36±2.51	77.68±2.09
	✓	×	×	79.85±1.80	81.85±2.30	78.20±1.95
	✓	✓	×	80.58±1.75	82.63±1.62	79.08±2.01
	✓	✓	✓	**81.85±0.76**	**83.42±1.31**	**80.02±0.87**

incorporating SADA_V3 improves spectral–spatial joint learning, providing additional 0.7%–1.4% OA improvements; and integrating LCSF enables cross-domain semantic consistency, further boosting OA by 0.8%–2.5%. These results demonstrate the plug-and-play capability of our modules and their generalizability to existing FSL frameworks for hyperspectral classification.

3) Parameter Analysis: The tunable parameters of MSS-CFSL primarily include the size of the initialized spatial knowledge base N_{spa}, the spectral knowledge base N_{spe}, and the high-level shared-feature set N_s. These parameters are responsible for enhancing the representation of low-level spatial–spectral style features and capturing high-level shared information across domains. To investigate the sensitivity of MSS-CFSL to these parameters, we conducted a comprehensive parameter analysis. The value ranges were set as follows: $N_{\text{spe}} \in \{10, 20, 30\}$, $N_{\text{spa}} \in \{10, 20, 30, 40, 50\}$, and $N_s \in \{10, 20, 30, 40, 50\}$. Experiments were performed on the four datasets to evaluate the impact of different parameter combinations on model performance.

As shown in Fig. 17, the parameter selection of MSS-CFSL significantly affects model performance and follows a clear optimization pattern. Specifically, the spectral knowledge base capacity (N_{spe}): a smaller value ($N_{\text{spe}} = 10$) yields the best performance on the IP dataset (OA = 79.41%), while an excessively large N_{spe} leads to spectral statistical redundancy and reduces classification accuracy (OA decreases by 1.5%–2.0%). This demonstrates that spectral feature modeling requires a tradeoff between information completeness and noise suppression. Collaboration between the spatial knowledge base and shared features (N_{spa}, N_s): both parameters remain stable in the lower range (10–30), with peak performance achieved when the ratio is close to 1:1 (e.g., $N_{\text{spa}} = 10$ and $N_s = 10$). When either parameter exceeds 40, OA drops significantly (e.g., OA = 77.73% for $N_{\text{spa}} = 50$ and $N_s = 50$), indicating that over-parameterization harms the migratability of cross-domain features. A similar phenomenon is presented on the datasets UP, Salians, and HO. Based on the experimental results, $N_{\text{spe}} = 10$, $N_{\text{spa}} = 10$, and $N_s = 10$ are set as the default parameters for the cross-domain small-sample task, achieving the highest OA (79.41%) and optimal computational efficiency on the IP dataset.

V. CONCLUSION

In this study, we introduced the adaptive feature integration for MSS-CFSL method to address the challenges of domain discrepancies and limited labeled samples in HSIC. By enhancing both low-level style features and high-level semantic features, our approach effectively captures critical information and improves performance through spatial–spectral feature fusion. Extensive experiments on several hyperspectral datasets show that MSS-CFSL outperforms existing state-of-the-art methods, achieving significant improvements in OA, AA, and the kappa coefficient. These results highlight the superior effectiveness of MSS-CFSL in CDFSL tasks. Future work will focus on extending this method to other image classification domains and evaluating its performance in more complex settings.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] T. Rajamani, P. Sevugan, and S. Ragupathi, "Automatic building footprint extraction and road detection from hyperspectral imagery," *J. Electron. Imag.*, vol. 32, no. 1, 2023, Art. no. 011005.

[3] S. Hajaj, A. El Harti, A. B. Pour, A. Jellouli, Z. Adiri, and M. Hashim, "A review on hyperspectral imagery application for lithological mapping and mineral prospecting: Machine learning techniques and future prospects," *Remote Sens. Appl., Soc. Environ.*, vol. 35, Aug. 2024, Art. no. 101218.

[4] R. Rajabi, A. Zehtabian, K. D. Singh, A. Tabatabaeenejad, P. Ghamisi, and S. Homayouni, "Editorial: Hyperspectral imaging in environmental monitoring and analysis," *Frontiers Environ. Sci.*, vol. 11, Jan. 2024, Art. no. 1353447.

[5] P. K. Sethy, C. Pandey, Y. K. Sahu, and S. K. Behera, "Hyperspectral imagery applications for precision agriculture–a systemic survey," *Multimedia Tools Appl.*, vol. 81, no. 2, pp. 3005–3038, Jan. 2022.

[6] H. Chen et al., "M^{3}FuNet: An unsupervised multivariate feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5513015.

[7] M. Zhang, L. Liu, Y. Jin, Z. Lei, Z. Wang, and L. Jiao, "Tree-shaped multiobjective evolutionary CNN for hyperspectral image classification," *Appl. Soft Comput.*, vol. 152, Feb. 2024, Art. no. 111176.

[8] Y. Ding, Z. Zhang, H. Hu, F. He, S. Cheng, and Y. Zhang, "Graph sample and aggregate-attention network for hyperspectral image classification," in *Graph Neural Network for Feature Extraction and Classification of Hyperspectral Remote Sensing Images*. Cham, Switzerland: Springer, 2024, pp. 29–41.

[9] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.

[10] M. Ahmad, M. Usama, A. M. Khan, S. Distefano, H. A. Altuwaijri, and M. Mazzara, "Spatial–spectral transformer with conditional position encoding for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[11] M. Ahmad, U. Ghous, M. Usama, and M. Mazzara, "WaveFormer: Spectral–spatial wavelet transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[12] Z. Shu, Y. Wang, and Z. Yu, "Dual attention transformer network for hyperspectral image classification," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107351.

[13] A. Farahani, S. Voghoei, K. Rasheed, and H. R. Arabnia, "A brief review of domain adaptation," in *Proc. Adv. Data Sci. Inf. Eng. (ICDATA IKE)*, 2021, pp. 877–894.

[14] M. Wang and W. Deng, "Deep visual domain adaptation: A survey," *Neurocomputing*, vol. 312, pp. 135–153, Oct. 2018.

[15] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[16] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[17] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[18] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.

[19] P. Bateni, R. Goyal, V. Masrani, F. Wood, and L. Sigal, "Improved few-shot visual classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14493–14502.

[20] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[21] H. Wang, X. Wang, and Y. Cheng, "Graph meta transfer network for heterogeneous few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501112.

[22] Y. Xu, Y. Zhang, T. Yue, C. Yu, and H. Li, "Graph-based domain adaptation few-shot learning for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 4, p. 1125, Feb. 2023.

[23] N. Li, D. Zhou, J. Shi, X. Zheng, T. Wu, and Z. Yang, "Graph-based deep multitask few-shot learning for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 9, p. 2246, May 2022.

[24] Q. Ran et al., "Deep transformer and few-shot learning for hyperspectral image classification," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1323–1336, Jan. 2023.

[25] Y. Peng, Y. Liu, B. Tu, and Y. Zhang, "Convolutional transformer-based few-shot learning for cross-domain hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1335–1349, 2023.

[26] C. Yu, C. Liu, H. Yu, M. Song, and C.-I. Chang, "Unsupervised domain adaptation with dense-based compaction for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12287–12299, 2021.

[27] X. Tang, C. Li, and Y. Peng, "Unsupervised joint adversarial domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536415.

[28] S. Zhang, M. Xu, J. Zhou, and S. Jia, "Unsupervised spatial–spectral CNN-based feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524617.

[29] B. Deng, S. Jia, and D. Shi, "Deep metric learning-based feature embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1422–1435, Feb. 2020.

[30] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] Z. Fang et al., "Confident learning-based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527116.

[32] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[33] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, Feb. 2024.

[34] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[35] B. Park, W. R. Windham, K. C. Lawrence, and D. P. Smith, "Contaminant classification of poultry hyperspectral imagery using a spectral angle mapper algorithm," *Biosyst. Eng.*, vol. 96, no. 3, pp. 323–333, Mar. 2007.

[36] J. Xia, P. Ghamisi, N. Yokoya, and A. Iwasaki, "Random forest ensembles and extended multiextinction profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 202–216, Jan. 2018.

[37] S. Xu, S. Liu, H. Wang, W. Chen, F. Zhang, and Z. Xiao, "A hyperspectral image classification approach based on feature fusion and multi-layered gradient boosting decision trees," *Entropy*, vol. 23, no. 1, p. 20, Dec. 2020.

[38] T. Moughal, "Hyperspectral image classification using support vector machine," in *Proc. J. Phys., Conf.*, 2013, vol. 439, no. 1, Art. no. 012042.

[39] H. Du, H. Qi, X. Wang, R. Ramanath, and W. E. Snyder, "Band selection using independent component analysis for hyperspectral image processing," in *Proc. 32nd Appl. Imag. Pattern Recognit. Workshop*, Oct. 2003, pp. 93–98.

[40] X. He and Y. Chen, "Modifications of the multi-layer perceptron for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 17, p. 3547, Sep. 2021.

[41] Z. Gong, L. Tong, J. Zhou, B. Qian, L. Duan, and C. Xiao, "Superpixel spectral–spatial feature fusion graph convolution network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536216.

[42] Y. Su et al., "ACGT-Net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.

[43] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.

[44] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[45] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[46] Z. Huang, H. Tang, Y. Li, and W. Xie, "HFC-SST: Improved spatial–spectral transformer for hyperspectral few-shot classification," *J. Appl. Remote Sens.*, vol. 17, no. 2, May 2023, Art. no. 026509.

[47] K.-K. Huang, H.-T. Yuan, C.-X. Ren, Y.-E. Hou, J.-L. Duan, and Z. Yang, "Hyperspectral image classification via cross-domain few-shot learning with kernel triplet loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530818.

[48] K. Zhou, Y. Yang, Y. Qiao, and T. Xiang, "Domain generalization with MixStyle," 2021, *arXiv:2104.02008*.

[49] W. Wang, L. Duan, Y. Wang, J. Fan, and Z. Zhang, "MMT: Cross domain few-shot learning via meta-memory transfer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15018–15035, Dec. 2023.

[50] S. Zhong, C.-I. Chang, and Y. Zhang, "Iterative support vector machine for hyperspectral image classification," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3309–3312.

[51] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[52] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[53] S. Jia, Y. Wang, S. Jiang, and R. He, "A center-masked transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510416.

[54] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[55] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.

[56] Z. Ye, J. Wang, T. Sun, J. Zhang, and W. Li, "Cross-domain few-shot learning based on graph convolution contrast for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504614.

[57] H. Liu, J. He, Y. Li, and Y. Bi, "Multilevel prototype alignment for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4400115.

[58] A. Qin et al., "Few-shot learning with prototype rectification for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409015.

[59] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:31:18 UTC from IEEE Xplore. Restrictions apply.